Forecast-to-Fulfilment

Probabilistic Demand Forecasting, Two-Stage Stochastic Capacity Planning and Online Order Assignment for an Iberian E-commerce Fulfilment Network

Panagiotis Kiousis

October 2026

Abstract

We study the forecast-to-fulfilment decision chain of an e-commerce network covering mainland Spain and Portugal: 65 demand zones, 10 fulfilment centres, 12 sort centres, four product categories and about 450,000 units per day. Demand is produced by a generative model with known ground truth, which lets us evaluate every decision against the *true* conditional demand distribution.

- **Forecasting.** A global LightGBM quantile model with event-aligned features and rolling conformal calibration halves the day-ahead error of a seasonal-naive baseline (wMAPE 11.9% vs. 24.3%) and outperforms a zero-shot time-series foundation model (Chronos-Bolt, 15.4%). A Gaussian copula on probability-integral-transform residuals turns the marginal quantiles into calibrated joint scenarios.
- **Planning.** Trucks and labour are planned the day before with a two-stage stochastic mixed-integer program whose recourse is the order-assignment LP. A Benders decomposition with an LP phase, in-out stabilisation, a Jensen-based partial decomposition and parallel warm-started subproblems solves 100-scenario instances (254,000 columns) to 0.05% in seconds: at least 90× faster than classic Benders and 12× faster than the extensive form solved by HiGHS, and it scales to 1000 scenarios (2.5 million columns) in 41 s. Sample average approximation bounds certify the plans.
- **Decision-focused evaluation.** The remaining forecast error costs €16k per day; a per-series P90 buffer costs €96k per day; the stochastic plan beats the deterministic one by a margin that grows sharply when same-day recourse is expensive.
- **Speed and emissions.** Exact cost–speed–CO_2 frontiers show that next-day delivery is cheap up to a network-imposed ceiling.
- **Online assignment.** LP-based fluid tracking closes 79% of the regret of greedy assignment with respect to a clairvoyant benchmark.
- **Investment.** The same models value a new fulfilment centre near Lisbon, which raises next-day delivery in Portugal from 54% to 93% and saves €9–€18M per year.

Contents

1 Introduction **3**

2 Related work **4**

3 Problem setting, network and data **5**
3.1 The fulfilment network 5
3.2 A generative demand model 6
3.3 Decision hierarchy 8

4 Probabilistic demand forecasting **8**
4.1 Protocol 8
4.2 Models 8

4.3 Results . . . 9
4.4 From marginal quantiles to joint scenarios . . . 10

5 Day-ahead capacity planning as a two-stage stochastic program 11
5.1 Formulation . . . 11
5.2 Structural properties . . . 12
5.3 Sample average approximation . . . 13
5.4 Benders decomposition . . . 13
5.5 Benchmarks for the planning paradigm . . . 14

6 Computational study 14

7 Planning results 16
7.1 Statistical quality of the SAA solutions . . . 16
7.2 When does stochastic planning pay? . . . 18
7.3 What is a better forecast worth? Decision-focused evaluation . . . 20

8 Multi-objective analysis: cost, speed and CO_2 21

9 Online order assignment under the day-ahead plan 22

10 Strategic what-if: a fulfilment centre near Lisbon 24

11 Discussion, limitations and next steps 26

A Notation 29

B Forecasting features 29

C Reproducibility 29

1 Introduction

Every evening, an e-commerce fulfilment organisation commits money it cannot easily take back. Line-haul trucks are tendered to carriers for the next night. Labour shifts are scheduled at every fulfilment centre. These commitments are made against a forecast; the orders they will serve have not been placed yet. The next day, millions of units arrive one by one and each must be assigned, irrevocably, to an FC that has the item, the labour and space on a truck, and ideally a lane short enough to keep the next-day promise. When the committed capacity is in the wrong place, the network pays for spot trucks, overtime, late deliveries and broken promises.

This report builds the complete chain of models behind these decisions for a realistically dimensioned network covering mainland Spain and Portugal: 65 demand zones, 10 FCs, 12 sort centres, 4 product categories and about 450,000 units per day, rising beyond a million on Black Friday and Prime Day. The aim is to treat the three layers, *forecast* → *plan* → *assign*, as one system and to evaluate each layer by what it costs the next one.

Executive summary.

- **Forecasting.** A global LightGBM quantile model with event-aligned features halves the day-ahead error of a seasonal-naive forecast (wMAPE 24.3% → 11.9%), and beats a zero-shot foundation model (Chronos-Bolt, 15.4%). Rolling conformal calibration fixes its overconfidence (80% intervals: 71.6% → 79.3% coverage). A Gaussian copula on forecast residuals makes the *national* uncertainty honest: the coverage of a 90% interval rises from 30% to 82%.
- **Planning.** The day-ahead truck and labour plan is a two-stage stochastic MIP. It has 130 integer variables and a recourse LP per scenario, i.e. 254,000 columns at 100 scenarios. Benders decomposition with an LP phase, in-out stabilisation and a partial-decomposition (Jensen) block solves it to 0.05% in seconds: at least 90× faster than classic Benders and 12× faster than the extensive form at 100 scenarios. Sample average approximation (SAA) bounds certify the solutions to within 0.4% of the true optimum (95% confidence).
- **Value of forecasts.** Scored on the true demand distribution over 31 test days, the remaining forecast error costs €16k per day (about €6M per year). A per-series P90 buffer, a common heuristic, wastes €96k per day. The stochastic plan beats the deterministic plan by €3–€10k per day when same-day recourse is cheap, and by €39k per day when it is expensive. In that regime copula scenarios are worth €20k per day more than independent ones, and the hedge flips from under- to over-provisioning exactly where newsvendor theory predicts.
- **Speed vs. cost vs. CO_2.** The Pareto frontier puts a price on the next-day promise. Raising the next-day share from 83.5% to 91% costs only €8k per day, after which the frontier turns vertical at a ceiling set by the network. Speed is bought by routing, not by extra capacity, and it raises line-haul CO_2 by 8%.
- **Online assignment.** Assigning orders greedily costs 7.1% more than a clairvoyant LP. LP-dual bid prices re-solved hourly close 67% of the gap, and fluid LP tracking closes 79%: €110k per day.
- **Investment what-if.** A fulfilment centre near Lisbon raises next-day delivery in northern Portugal by up to 85 percentage points, cuts CO_2 by 10% and saves €9.3M per year in operating cost (€18.3M including the value of speed).

Contributions.

1. An end-to-end, reproducible forecast-to-fulfilment pipeline: a generative demand model with known ground truth, probabilistic forecasting with conformal calibration and copula scenarios, a two-stage stochastic capacity planner, and an online LP-based controller, all evaluated in one consistent cost model.

2. A Benders implementation for the planning problem that combines warm-started parallel subproblems, an LP phase, in-out stabilisation, partial decomposition justified by a Jensen bound (Proposition 2) and an adaptive master tolerance. An ablation study isolates the contribution of each component (Section 6).
3. A *decision-focused* evaluation of forecasts. Because the data-generating process is known, every plan is scored against the true conditional demand distribution. This prices forecast errors, buffer heuristics and dependence modelling in euros per day, and shows *when* each matters.
4. Multi-objective and strategic analyses with the same machinery: exact cost–speed and cost–CO_2 frontiers (including unsupported Pareto points that a weighted sum misses), and the business case of a new fulfilment centre.

Outline.
- Section 2 reviews related work.
- Section 3 describes the network, the costs and the demand model.
- Section 4 develops the forecasts and the scenario generator.
- Section 5 formulates the stochastic planning problem and the decomposition, which Section 6 benchmarks computationally.
- Section 7 reports the planning results: statistical quality, value of the stochastic solution, value of forecasts.
- Section 8 covers the multi-objective analysis, Section 9 the online assignment, and Section 10 the network-design what-if.
- Section 11 discusses limitations and next steps.

All code, data generation and experiments are available and reproducible from a single `make` target (Appendix C).

2 Related work

E-commerce fulfilment decisions. Xu et al. [36] showed that periodically re-evaluating order-to-FC assignments made in real time yields substantial savings in an online retailer. Acimovic and Graves [1] proposed scoring each fulfilment option by its immediate shipping cost plus an estimate of future costs. They derived those estimates from the dual values of a transportation LP; on industry data the heuristic captured about a third of the gap to a clairvoyant benchmark. Jasin and Sinha [15] studied multi-item orders and introduced an LP-based correlated rounding scheme, later improved by Ma [19]. Our online layer (Section 9) follows this LP-dual tradition. It is closely related to bid-price controls in network revenue management [32], where shadow prices of capacity ration a perishable resource among sequentially arriving requests. We embed it in a two-level model in which the capacities themselves are decided the day before under uncertainty.

Two-stage stochastic programming and Benders decomposition. Capacity committed before demand is revealed is the textbook setting for two-stage stochastic programming [5, 31]. Sample average approximation with statistical optimality bounds is analysed by Kleywegt et al. [17] and Mak et al. [21]. Benders decomposition [4], known as the L-shaped method in this context [33], is the standard solution approach; Rahmaniani et al. [27] survey its many accelerations. We combine several of them:
- multi-cut [6];
- an LP phase [25];
- in-out stabilisation [3], popularised for large facility-location problems by Fischetti et al. [11];
- partial decomposition with artificial mean-value scenarios [8].

Pareto-optimal cuts [20] are a natural further extension.

Probabilistic demand forecasting. Global gradient-boosted tree models dominated the M5 competition [22], one of the largest public retail forecasting benchmarks. Amazon's own forecasting research produced the DeepAR [29] and MQ-RNN [35] families, and more recently the Chronos time-series foundation models [2]. Quantile forecasts are evaluated with proper scoring rules [13]: the quantile loss for marginals, and the energy and variogram scores [30] for joint distributions. Conformal prediction [28, 34] provides distribution-free calibration; under distribution shift it is typically applied in rolling or adaptive form [12]. We use a Gaussian copula [26] with shrinkage estimation [18] to turn marginal quantiles into joint scenarios.

From predictions to decisions. The decision-focused learning literature, e.g. the "smart predict-then-optimise" framework of Elmachtoub and Grigas [10], argues that forecasts should be judged by the quality of the decisions they induce. Our generative setting lets us measure this directly: every plan is scored by its expected cost under the *true* conditional demand distribution (Section 7.3).

Multi-objective optimisation. Weighted-sum scalarisation can only reach the *supported* points of a Pareto frontier [9]. The (augmented) ε-constraint method [23, 24] also recovers the unsupported points, which exist whenever integer variables make the frontier non-convex, as they do with integer trucks.

3 Problem setting, network and data

3.1 The fulfilment network

We study a stylised but realistically dimensioned outbound network for mainland Spain and Portugal (Figure 1). Demand originates in $|\mathcal{J}| = 65$ *zones*: the 47 mainland Spanish provinces and the 18 mainland Portuguese districts, located at their capitals and weighted by census population (53.9 million inhabitants in total). Units are shipped from $|\mathcal{I}| = 10$ fulfilment centres (FCs) located in well-known Iberian logistics hubs (two in Madrid, two in Barcelona, Zaragoza, Seville, Valencia, Asturias, Bilbao and Murcia), travel on scheduled line-haul trucks to one of $|\mathcal{H}| = 12$ sort centres (SCs), and are delivered by the last mile to the customer zone. A zone is served by the nearest SC of its own country, so the decision *which FC ships a unit* also fixes its middle-mile lane $(i, h(j))$. Products belong to $|\mathcal{K}| = 4$ categories with different volume v_k (truck package-equivalents per unit) and labour content l_k (labour units per unit): small sortable ($v = 1$, $l = 1$), medium sortable (2.2, 1.4), bulky non-sortable (7, 3) and apparel (1.3, 0.9). Bulky items are only stocked in the four large non-sortable sites, which makes their assignment far less flexible.

Distances, transit times and the speed promise. Road distances are great-circle distances inflated by a circuity factor of 1.22, which is typical for the Iberian road network. A lane's transit time is $\mathrm{km}/80 + 0.5$ hours (driving plus loading/unloading). Trucks leave the FC after the evening cut-off and must reach the SC before the early-morning sort. A unit is therefore delivered *on time* (next day) if and only if its lane fits the 6.5-hour overnight window, and *late* (day +2) otherwise. Only 48 of the 120 lanes are next-day feasible (Figure 1b); 23 zones (mostly Galicia, Portugal and Andalusia) can be served on time by exactly one FC, and 63% of all FC–zone–category arcs are late. This gives the cost–speed trade-off its bite: the cheapest FC with free capacity is often a late one.

Costs. All monetary values are in euros per day; Table 1 lists every parameter. A scheduled truck on lane (i, h) costs $F_{ih} = 180 + 1.25 \cdot \mathrm{km}_{ih}$ (€181–€1719) and carries $Q = 1400$ package-equivalents. Capacity bought on the day from the spot/LTL market costs 1.8× the scheduled rate per package-equivalent. Labour is scheduled in crews of 10 associates for one shift ($\pi = 2000$ labour units for €1760, i.e. €0.88 per labour unit). Up to 30% overtime can be added at a 1.6×

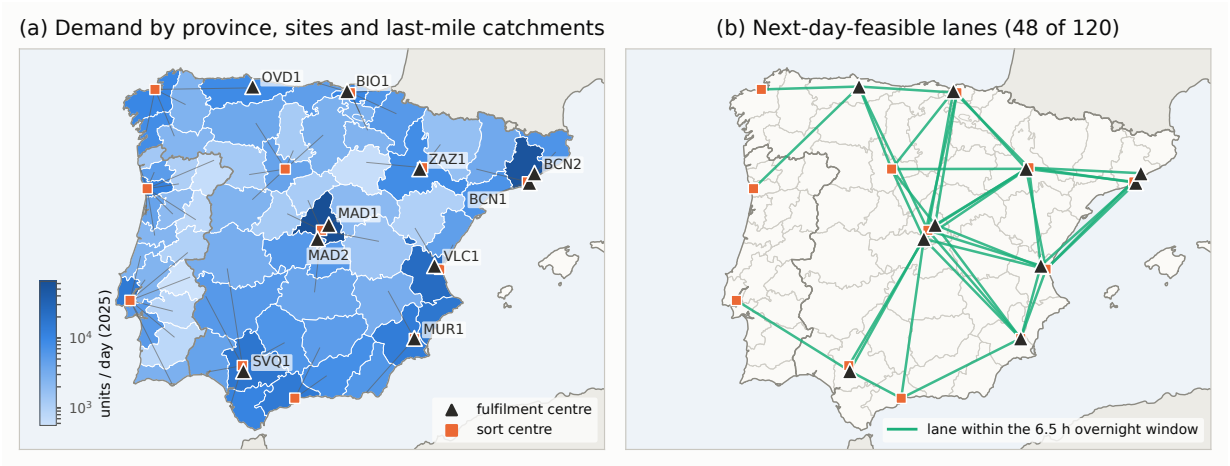


Figure 1: (a) The network on the Iberian Peninsula: provinces and districts shaded by mean daily demand (log scale), fulfilment centres (triangles), sort centres (squares) and the last-mile catchment of every sort centre (thin lines). (b) Lanes whose line-haul transit fits the overnight window, i.e. lanes that keep the next-day promise. Only 48 of the 120 lanes are next-day feasible, so where a unit ships from decides whether the promise is kept. Boundaries: Natural Earth (public domain).

Table 1: Network and cost parameters (base case).

Parameter	Value	Meaning
Q	1400 pkg-eq	trailer capacity
F_{ih}	$180 + 1.25\,\mathrm{km}_{ih}$	cost of a scheduled truck on lane (i, h)
s_{ih}	$1.8\,F_{ih}/Q$	spot capacity cost per package-equivalent
π, L	2000 labour units, €1760	capacity and cost of one crew-shift
ρ, c^{ot}	0.30, $1.6\,L/\pi = €1.41$	overtime cap (fraction) and cost per labour unit
W_i, D_i	22–70 crews, 60–150 trucks	site limits (total 426 crews, 1020 departures)
c_i^{fc}	€0.24–€0.31	FC variable processing cost per unit
c_j^{lm}	$2.60 + 0.012\,\mathrm{km}_{h(j),j}$	last-mile cost per unit
p, λ	€15, €1	penalty for an unshipped unit; price of a late unit
window	6.5 h at 80 km/h + 0.5 h	overnight line-haul window
CO_2	0.9 kg/truck-km; 0.18 kg/unit last mile	emission factors

premium. Each FC has a maximum number of crews and of outbound truck departures (dock slots). Shipping a unit incurs an FC processing cost and a last-mile cost ($2.60 + 0.012$ €/km from the SC, €2.61–€6.02). A unit that cannot be shipped today misses its promise and costs $p = €15$ (re-handling, compensation and cancellation risk; Section 7.2 discusses this choice). A late unit costs $\lambda = €1$, the *price of speed*; this weight is the dial of the multi-objective analysis in Section 8. CO_2 is accounted at 0.9 kg per truck-km, with a 1.35× factor for less consolidated spot capacity.

Inventory. Inventory placement is decided upstream on a longer horizon and is exogenous here. For each day, FC i holds Inv_{ik} units of category k: a national cover of 1.35 days of expected demand, split across the FCs stocking k in proportion to their size and distorted by a persistent log-normal imbalance (standard deviation 0.25). Inventory is thus not perfectly aligned with demand, which is what makes the *online* assignment problem (Section 9) non-trivial.

3.2 A generative demand model

Real order data are proprietary, so we build a *generative* data set. This is a feature, not only a necessity: because the latent states are known, the *true* conditional distribution of tomorrow's

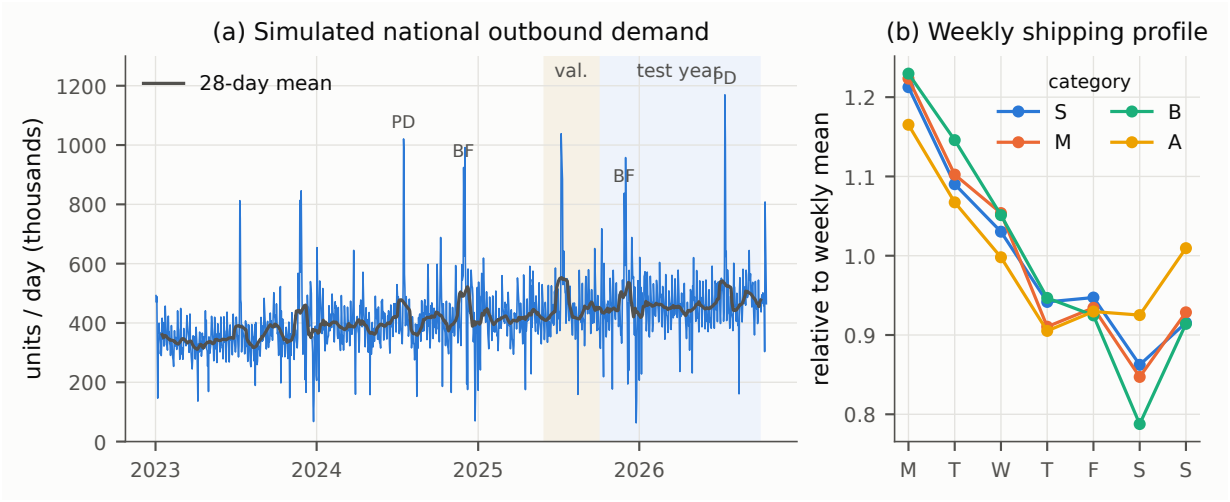


Figure 2: (a) Simulated national daily units to ship; shaded: validation window and test year (BF: Black Friday, PD: Prime Day). (b) Weekly profile per category: orders placed at the weekend ship on Monday; apparel is flatter.

demand can be sampled exactly. That provides an oracle forecaster, and it lets us measure the economic value of forecast quality without the noise of a single realised path (Section 7.3). Daily outbound units y_{tjk} for day t, zone j and category k follow

$$\begin{aligned} y_{tjk} &\sim \mathrm{NegBin}\big(\text{mean } \mu_{tjk},\ \text{dispersion } r_k\big), \\ \log \mu_{tjk} &= \underbrace{\log b_{jk} + \tau_{tj} + \omega_k(\mathrm{dow}_t) + \alpha_{k,c(j)}(\mathrm{doy}_t) + e_{tk,c(j)} + \eta^{\mathrm{hol}}_{tj} + \phi_{tk} + \beta_k^\top z_{tj}}_{\text{known at forecast time: calendar, promotions, weather}} \\ &\quad + \underbrace{\eta_t + \xi_{t,r(j)} + \zeta_{tk}}_{\text{latent AR(1) shocks}}. \end{aligned} \tag{1}$$

The terms are:

- a base level proportional to population, with an urbanisation premium, a lower e-commerce penetration in Portugal (0.72) and zone-specific category mixes;
- log-linear growth of 10% (Spain) and 16% (Portugal) per year;
- category-specific weekly profiles (units to *ship* peak on Monday, Figure 2b);
- a smooth annual cycle with a pronounced August dip in Spain;
- retail events: Black Friday week and Cyber Monday with a multi-day shipping tail, two Prime events per year, the Christmas ramp and trough, the Spanish *Reyes Magos* peak, the January *rebajas* for apparel, and back-to-school;
- national and regional public holidays (including moving Easter holidays), with a catch-up effect the day after;
- announced flash promotions per category;
- weather covariates z_{tj}: spatially correlated temperature anomalies and rain, which raise online demand slightly; cold snaps lift apparel.

The unpredictable part consists of three AR(1) processes: a national shock η_t (autocorrelation 0.8), regional shocks $\xi_{t,r}$ (0.6) and category shocks ζ_{tk} (0.7). On top of these comes negative-binomial count noise, with dispersion between 45 (bulky) and 160 (small sortable). The resulting series have a coefficient of variation of about 14% at the zone–category level and 5% for the national total.

The simulation covers January 2023 to December 2026. Volume averages 443 thousand units per day in 2025 and peaks at 1.17 million on Prime Day 2026 (Figure 2a); the scale is plausible for a large Iberian e-commerce network.

3.3 Decision hierarchy

The study follows the decision hierarchy of an outbound fulfilment organisation:

1. **Forecasting** (Section 4): every evening, probabilistic forecasts of next-week demand per zone and category, and joint scenarios for the next day.
2. **Day-ahead capacity planning** (Section 5–Section 8): commit the number of scheduled trucks per lane and labour crews per FC before demand is known, anticipating that the order assignment will adapt to realised demand.
3. **Online order assignment** (Section 9): during the day, assign each arriving unit to an FC irrevocably, consuming the committed capacity.
4. **Strategic what-if** (Section 10): evaluate a network investment with the same machinery.

4 Probabilistic demand forecasting

Capacity planning needs a *distribution* of tomorrow's demand at the zone–category level, and the joint behaviour of the 260 series. This section builds the marginal forecasts; Section 4.4 turns them into joint scenarios.

4.1 Protocol

We use a rolling-origin backtest. Every day τ we forecast $h = 1, \dots, 7$ days ahead for all 260 series, using only information available at the end of day τ. Models are retrained quarterly on an expanding window, on six dates from June 2025 to September 2026; no training target lies after the retraining date. The origins are split in two:

- a **validation** period (June–September 2025, including Prime Day 2025), used for every modelling decision;
- a **test year** (targets from 1 October 2025 to 30 September 2026), never used for design decisions.

Accuracy is measured by:

- the weighted MAPE (wMAPE) of the median, $\sum |y - \hat{y}_{0.5}| / \sum y$, and its bias;
- the scaled quantile loss (SQL) over seven levels $a \in \{0.05, 0.1, 0.25, 0.5, 0.75, 0.9, 0.95\}$, $\text{SQL} = \frac{1}{7} \sum_a \sum \rho_a(y - \hat{y}_a) / \sum y$, where ρ_a is the pinball loss. Twice the SQL approximates the scaled CRPS, a strictly proper score [13];
- the empirical coverage and width of the central 80% and 90% intervals.

4.2 Models

- **Seasonal naive**: $\hat{y}_{\tau+h} = y_{\tau+h-7}$. Its quantiles come from the trailing 8-week empirical distribution of multiplicative errors, per series and horizon.
- **LightGBM-Q** [16]: one *global* gradient-boosted model per quantile level, trained with the pinball loss on all series and horizons jointly (1.7 million rows per retraining). The target is $y_{\tau+h}/\bar{y}_\tau$, normalised by the trailing 28-day mean, which makes the problem scale-free: one model serves Soria and Madrid. There are 43 features (Appendix B): lags, momentum at the series, regional and national level, the calendar of the target day, announced promotions, weather forecasts, and *event-aligned* year-over-year ratios. The latter compare with last year's Black Friday or Prime Day at the same offset, rather than with the same calendar date. Quantile crossing is removed by monotone rearrangement [7].
- **Chronos-Bolt** [2]: a pretrained time-series foundation model (small, 48M parameters, and base, 205M), applied *zero-shot* to the raw history (context up to 2048 days) without covariates. It outputs the 0.1–0.9 quantiles; the outer levels are extrapolated in normal-score space. Inference over all 487 origins and 260 series takes under three minutes on one consumer GPU (RTX 5070 Ti).

Table 2: Forecast accuracy on the test year (260 series, 365 origins, all quantities aggregated over series). SQL: scaled quantile loss; width: mean 80% interval width relative to demand; CP: conformal calibration.

	wMAPE (%)		Bias (%)	SQL (%)		Coverage (%)		Width
Method	h=1	h=7	h=1	h=1	h=7	80%	90%	80%
Seasonal naive	24.3	24.6	+0.4	8.02	8.20	77.6	86.6	79.8
Seasonal naive + CP	24.6	25.1	+0.7	8.32	8.81	76.8	86.4	89.2
Chronos-Bolt-S	15.5	16.9	-2.3	4.80	5.30	79.3	88.0	45.6
Chronos-Bolt-S + CP	15.5	17.2	-1.3	4.84	5.47	79.5	89.4	46.8
Chronos-Bolt-B	15.4	16.8	-1.9	4.75	5.27	81.0	89.2	48.0
Chronos-Bolt-B + CP	15.4	17.2	-1.2	4.79	5.42	79.5	89.3	46.7
LightGBM-Q	11.8	13.0	-0.6	3.46	3.79	71.6	83.1	31.0
LightGBM-Q + CP	11.9	13.3	-0.4	3.42	3.81	79.3	89.5	37.5
Oracle	9.6	11.1	-0.2	2.74	3.18	80.2	90.2	31.2

- **Conformal calibration (+CP).** Every forecaster can be recalibrated by rolling split-conformal prediction, separately for each quantile level a, horizon h and category k. The corrected quantile is

$$\tilde{y}_a = \hat{y}_a + \hat{y}_{0.5} \cdot \text{Quantile}_a\left\{\frac{y_{o'+h} - \hat{y}_{a,o'}}{\hat{y}_{0.5,o'}}\right\},$$

  where the scores come from the last 56 origins o' whose outcome is already known at the current origin. This is the per-level form of conformalised quantile regression [28]. A rolling window adapts to drift, in the spirit of adaptive conformal inference [12].
- **Oracle**: Monte Carlo quantiles of the true conditional distribution (Section 3.2). The oracle knows the latent states at the origin and the realised weather; it marks the irreducible noise floor.

Model selection on the validation period. Two choices were made on validation data only.
- Replacing the day-of-year Fourier terms by the month reduced the mean validation wMAPE from 12.61% to 12.38%. With fine-grained seasonal encodings and only two past years, the trees memorise dates instead of learning event effects.
- The event-aligned year-over-year features were adopted. Larger models (1500 rounds, 127–255 leaves) did not help.

4.3 Results

Table 2 and Figure 3 summarise the test year.

1. **The global LightGBM model halves the error of the seasonal naive benchmark**: day-ahead wMAPE falls from 24.3% to 11.8%, and quantile loss from 8.0% to 3.5%. It closes about 85% of the gap between the naive forecast and the oracle (9.6%). The remaining gap is concentrated around retail events, which occur only twice in the training history.
2. **Raw quantile regression is overconfident.** Its 80% interval covers only 71.6% of outcomes, a common failure of quantile GBMs on drifting data. Rolling conformal calibration restores nominal coverage (79.3% and 89.5%) at a negligible cost in accuracy, by widening the intervals from 31% to 38% of demand.
3. **The foundation model is a strong zero-shot baseline.** Chronos-Bolt is well calibrated out of the box and, without any training on this data, cuts the naive error by a third (wMAPE 15.4%). It still trails the covariate-aware LightGBM model by 3.5 points. The reason is visible in Figure 2: much of the predictable variance is driven by calendar events, holidays and

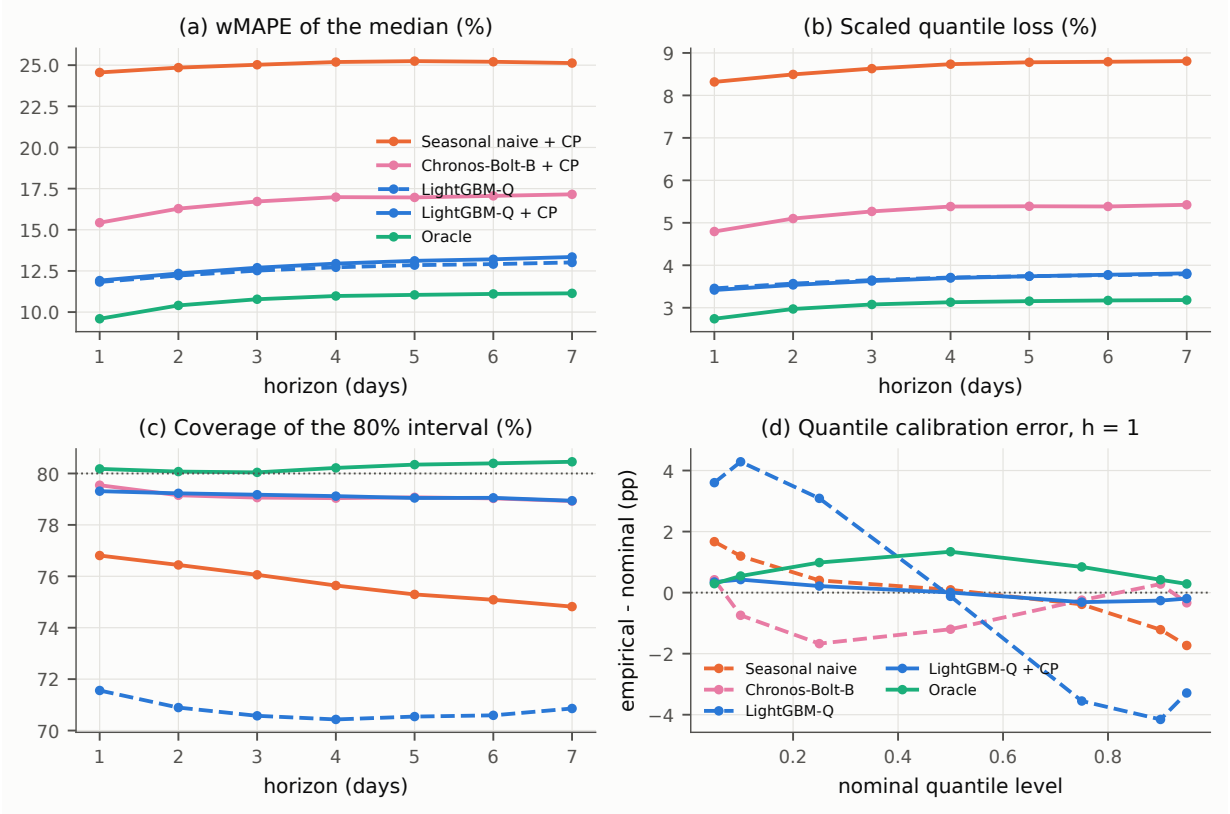


Figure 3: (a)–(c) Test-year accuracy by horizon. (d) Calibration error of each quantile level at $h = 1$ (empirical frequency of $y \leq \hat{y}_a$ minus a): raw LightGBM-Q is overconfident in both tails, and conformal calibration removes the error.

promotions that a history-only model cannot anticipate. Fine-tuning with covariates is the natural next step (Section 11).

Feature importance (gain, averaged over the six retrainings) confirms this reading. Weekday (17%) and holiday status (16%) dominate, followed by month, the 8-week same-weekday mean, the event-aligned year-over-year ratio and the announced promotion lift (7% each). These are all information a history-only model does not have.

4.4 From marginal quantiles to joint scenarios

The planning problem is driven by *aggregates*: the load on a lane is a sum over zones and categories, and the labour of an FC a sum over everything it ships. How uncertainty adds up therefore matters as much as the marginals. We generate joint scenarios with a Gaussian copula [26].

1. **Marginals.** The quantile function of every series is interpolated piecewise-linearly in normal-score space $z = \Phi^{-1}(a)$ through the seven forecast quantiles. Beyond the outer levels it is extrapolated linearly, which gives a locally Gaussian tail, and it is truncated at zero.
2. **Dependence.** For the last 120 days we compute the probability integral transform (PIT) of each realised value under its own forecast and map it to a normal score. The correlation of these scores across the 260 series, estimated with Ledoit–Wolf shrinkage [18], defines the copula.
3. **Sampling.** Draw $\varepsilon \sim \mathcal{N}(0, \hat{\Sigma})$ and map each coordinate through its quantile function.

The PIT residuals contain whatever the forecaster failed to predict: national and regional shocks, weather systems, category promotions. Their correlation is precisely the dependence that remains *conditionally* on the forecast.

Table 3 and Figure 4 show why this step matters. With independent sampling the individual

Table 3: Quality of joint day-ahead scenarios on the test year (every third day, 1000 scenarios). Energy and variogram scores are computed on the 48 SC×category totals (thousands of units) that load the network; lower is better. Coverage refers to the national daily total.

Scenario generator	Energy score	Variogram score	National PI coverage (%) 50%	90%	National CV (%)
Seasonal naive + CP, independent	18.72	1068	16	34	5.4
Seasonal naive + CP, copula	17.53	1013	60	91	25.7
LightGBM-Q + CP, independent	8.23	241	13	30	1.9
LightGBM-Q + CP, copula	8.03	242	40	82	6.9
True conditional distribution	5.88	162	52	88	4.8

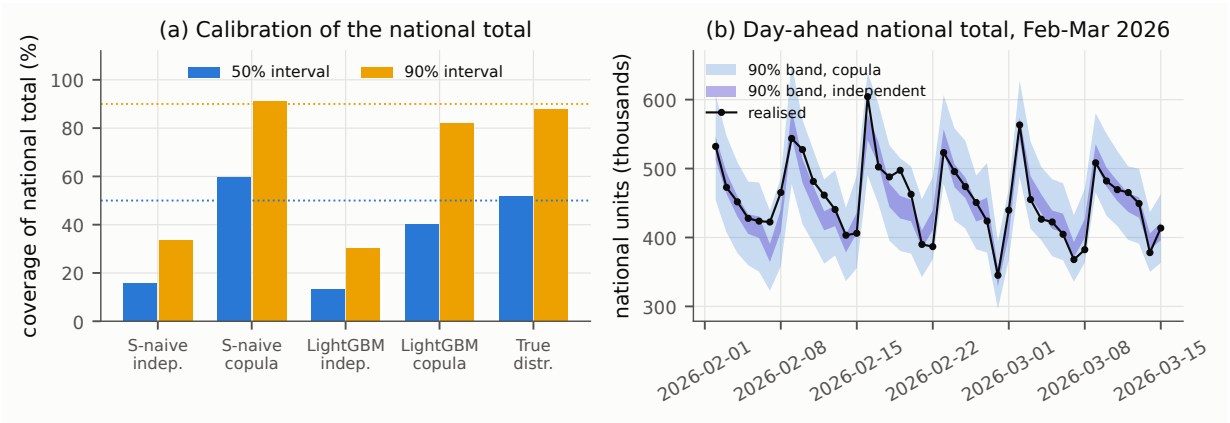


Figure 4: (a) Coverage of the national-total prediction intervals implied by the scenarios. Treating the 260 series as independent makes the national total look three times more certain than it is. (b) Day-ahead 90% bands for the national total: independent sampling (dark) is far too narrow; the copula (light) contains the realised values.

series are well calibrated, but the national total is not: its 90% interval covers the realised value only 30% of the time, because the coefficient of variation shrinks from 6.9% to 1.9% when correlated errors are wrongly assumed to cancel. The copula restores the coverage to 82% (50% interval: 40%) and improves the energy score on the SC–category totals. The residual under-coverage reflects a correlation estimated from 120 noisy PIT vectors with shrinkage towards independence, which slightly underestimates the common shock. The copula intervals are wider than the oracle's (CV 6.9% vs. 4.8%), as they should be: the forecaster cannot observe the latent states the oracle sees.

5 Day-ahead capacity planning as a two-stage stochastic program

5.1 Formulation

Every evening the planner commits two kinds of capacity for the next day before demand is known:

- scheduled line-haul trucks $y_{ih} \in \mathbb{Z}_+$ on every lane;
- labour crews $w_i \in \mathbb{Z}_+$ at every FC.

Both are cheap when committed in advance and expensive to adjust on the day. Once demand $\mathbf{d}$ is revealed, units are assigned to FCs, and overtime (o_i), spot capacity (z_{ih}) and unshipped units (u_{jk}) are the recourse actions. Let $\mathcal{A} = \{(i, j, k) : \text{FC } i \text{ stocks } k\}$ be the 2145 assignment

arcs, $h(j)$ the SC of zone j, and $\ell_{ij} \in \{0,1\}$ the late indicator of lane $(i, h(j))$. The **first stage** is

$$\begin{aligned} \min_{\mathbf{y},\mathbf{w}} \quad & \sum_{i,h} F_{ih}\, y_{ih} + L \sum_{i} w_i + \mathbb{E}_{\mathbf{d}}\big[\, Q(\mathbf{y}, \mathbf{w}, \mathbf{d}) \,\big] \\ \text{s.t.} \quad & \textstyle\sum_h y_{ih} \le D_i, \quad w_i \le W_i \quad \forall i, \qquad \mathbf{y} \in \mathbb{Z}_+^{\mathcal{I}\times\mathcal{H}},\ \mathbf{w} \in \mathbb{Z}_+^{\mathcal{I}}, \end{aligned} \tag{2}$$

and the **recourse** (second stage) is the linear program

$$\begin{aligned} Q(\mathbf{y}, \mathbf{w}, \mathbf{d}) = \min\ & \textstyle\sum_{(i,j,k)\in\mathcal{A}} \big(c^{\mathrm{fc}}_i + c^{\mathrm{lm}}_j + \lambda\, \ell_{ij}\big) x_{ijk} \\ & + c^{\mathrm{ot}} \textstyle\sum_i o_i + \sum_{i,h} s_{ih} z_{ih} + (p + \lambda) \sum_{j,k} u_{jk} && && (3) \\ \text{s.t.}\ & \textstyle\sum_j x_{ijk} \le \mathrm{Inv}_{ik} && \forall i,k \quad [\text{inventory}] && (4) \\ & \textstyle\sum_{j,k} l_k x_{ijk} - o_i \le \pi\, w_i && \forall i \quad [\text{regular labour}] && (5) \\ & o_i \le \rho\, \pi\, w_i && \forall i \quad [\text{overtime cap}] && (6) \\ & \textstyle\sum_{j:h(j)=h} \sum_k v_k x_{ijk} - z_{ih} \le Q\, y_{ih} && \forall i,h \quad [\text{truck capacity}] && (7) \\ & \textstyle\sum_i x_{ijk} + u_{jk} = d_{jk} && \forall j,k \quad [\text{demand}] && (8) \\ & x, o, z, u \ge 0. \end{aligned}$$

The recourse is a *capacitated transportation problem with side constraints* (433 rows and 2535 columns per scenario). It is the same LP that, with remaining capacities and remaining expected demand, drives the online bid prices of Section 9. Two features make the first-stage decision a genuinely stochastic one:

- scheduled capacity is cheaper than its flexible substitutes (spot is 1.8× and overtime 1.6× the scheduled rate), but it is bought before demand is known;
- the network can *pool* risk by re-routing units between FCs, at the price of lateness, so the right amount of capacity depends on the joint distribution of demand, not only on its marginals.

5.2 Structural properties

Write the recourse compactly as $Q(\mathbf{y}, \mathbf{w}, \mathbf{d}) = \min\{c^\top x : Ax \le b(\mathbf{y}, \mathbf{w}, \mathbf{d})\}$, where the right-hand side b is affine in $(\mathbf{y}, \mathbf{w}, \mathbf{d})$ and the demand rows are equalities.

Proposition 1 (Complete recourse and convexity)**.** *For every $(\mathbf{y}, \mathbf{w}) \ge 0$ and $\mathbf{d} \ge 0$ the recourse LP is feasible and bounded. Q is jointly convex and piecewise linear in $(\mathbf{y}, \mathbf{w}, \mathbf{d})$. With $\pi^\star$ an optimal dual vector, the vector*

$$g_{y,ih} = Q\, \pi^\star_{\mathrm{lane},ih}, \qquad g_{w,i} = \pi\, (\pi^\star_{\mathrm{lab},i} + \rho\, \pi^\star_{\mathrm{ot},i}) \tag{9}$$

is a subgradient of $Q(\cdot, \cdot, \mathbf{d})$ at $(\mathbf{y}, \mathbf{w})$, and $g_y, g_w \le 0$.

Proof. $(x, o, z, u) = (0, 0, 0, \mathbf{d})$ is feasible, and all costs are non-negative, so the LP is feasible and bounded below by 0. By LP duality, $Q = \max\{b(\mathbf{y}, \mathbf{w}, \mathbf{d})^\top \pi : \pi \in \Pi\}$, where the dual feasible set Π does not depend on $(\mathbf{y}, \mathbf{w}, \mathbf{d})$. A pointwise maximum of affine functions of an affine argument is convex and piecewise linear. For an optimal $\pi^\star$ and any other $(\mathbf{y}', \mathbf{w}')$, dual feasibility gives $Q(\mathbf{y}', \mathbf{w}', \mathbf{d}) \ge b(\mathbf{y}', \mathbf{w}', \mathbf{d})^\top \pi^\star = Q(\mathbf{y}, \mathbf{w}, \mathbf{d}) + \pi^{\star\top}(b' - b)$. Since b is affine with $\partial b / \partial y_{ih} = Q$ on the lane row, and $\partial b / \partial w_i = \pi$ and $\rho\pi$ on the labour and overtime rows, the result follows. Duals of $\le$ rows are non-positive in a minimisation, hence the sign. □

Proposition 2 (Jensen bound for scenario groups)**.** *Let $C \subseteq \mathcal{S}$ be a group of scenarios with total probability P_C and mean demand $\bar{\mathbf{d}}_C = \sum_{s\in C} (p_s / P_C)\, \mathbf{d}^s$. Then for all $(\mathbf{y}, \mathbf{w})$*

$$\textstyle\sum_{s\in C} p_s\, Q(\mathbf{y}, \mathbf{w}, \mathbf{d}^s) \ \ge\ P_C\, Q(\mathbf{y}, \mathbf{w}, \bar{\mathbf{d}}_C).$$

Proof. Immediate from convexity of Q in $\mathbf{d}$ (Proposition 1) and Jensen's inequality. □

The Jensen bound is what makes *partial* Benders decomposition [8] valid here. We keep a copy of the recourse LP of the mean demand of each scenario group *inside* the master problem and link it to the scenario variables θ_s. The master then "knows" from the very first iteration how capacity is consumed by demand. Without it, the master only learns this through cuts, one at a time.

5.3 Sample average approximation

The expectation in (2) is replaced by an average over N scenarios $\mathbf{d}^1, \ldots, \mathbf{d}^N$, drawn from the forecast distribution of Section 4.4; this is sample average approximation (SAA, [17, 31]). The SAA is a large mixed-integer program. Its deterministic equivalent (extensive form) has 130 integer and $2535N$ continuous variables and $433N + 10$ rows, so for $N = 100$ it has a quarter of a million columns. Section 7.1 quantifies the statistical quality of the SAA solutions with the Monte Carlo bounds of Mak et al. [21].

5.4 Benders decomposition

The SAA is solved by Benders decomposition [4, 33]. The **master problem** keeps the first-stage variables and one epigraph variable θ_s per scenario (multi-cut, [6]):

$$\begin{aligned} \min\ & F^\top \mathbf{y} + L\,\mathbf{1}^\top \mathbf{w} + \textstyle\sum_s p_s \theta_s \\ \text{s.t.}\ & \theta_s \geq Q_s(\hat{\mathbf{y}}, \hat{\mathbf{w}}) + g_y^{s\top}(\mathbf{y} - \hat{\mathbf{y}}) + g_w^{s\top}(\mathbf{w} - \hat{\mathbf{w}}) \quad \text{for every generated cut,} \\ & \theta_s \geq \underline{\theta}_s, \quad \text{site limits,} \quad \mathbf{y}, \mathbf{w} \text{ integer,} \end{aligned} \tag{10}$$

where $\underline{\theta}_s = \sum_{j,k} d^s_{jk} \min\{\min_i c_{ijk},\, p + \lambda\}$ is a trivial bound: every unit is either shipped at no less than its cheapest arc cost, or left unshipped.

Implementation and enhancements. Each of the following is evaluated in the ablation study of Section 6.

1. **Warm-started subproblems.** Each worker process owns a single HiGHS LP model [14] and swaps scenarios in and out. Scenarios and first-stage values only change row bounds, so dual simplex re-optimises from the previous optimal basis. A subproblem typically takes about 5 ms.
2. **Parallel subproblems.** Scenarios are partitioned across persistent worker processes.
3. **LP phase** [25]. The master's integrality is first relaxed and Benders runs to convergence on the LP relaxation. The cuts are valid for the integer master, so phase 2 starts with a good outer approximation.
4. **In-out stabilisation** [3, 11]. In the LP phase, cuts are separated at $\tilde{\mathbf{y}} = \alpha\hat{\mathbf{y}} + (1 - \alpha)\bar{\mathbf{y}}$, a convex combination of the master solution and a stability centre that is updated to $\tilde{\mathbf{y}}$; we use $\alpha = 0.5$. This damps the zig-zagging of Kelley's cutting-plane method. When no cut is violated at $\tilde{\mathbf{y}}$, we separate at $\hat{\mathbf{y}}$ directly.
5. **Partial decomposition (Jensen blocks).** Scenarios are sorted by national volume and split into G groups. The recourse LP of each group mean is added to the master together with the row $\sum_{s \in C} p_s \theta_s \geq P_C\, c^\top x^C$ (Proposition 2). By default ($G = 1$) the block is only active in the LP phase. In the integer phase it would make each master MIP heavier, and the cuts it helped to generate are kept anyway.
6. **Adaptive master tolerance.** In the integer phase the master is solved to a relative gap $\epsilon_k = \max\{\epsilon/4, \min(\epsilon_{k-1}, \mathrm{gap}_k/4)\}$, and the master's *dual* bound is used as the global lower bound. The master is thus solved only as accurately as the current Benders gap requires, without invalidating the bound. The incumbent is passed as a MIP start.

Complete recourse means every master iterate is feasible, so every iterate yields a valid upper bound and only optimality cuts are needed. The algorithm stops at a relative gap of $\epsilon = 5 \cdot 10^{-4}$. This is deliberately not tighter: at $N = 100$ the SAA objective itself has a standard error of about 0.1% (Section 7.1), so additional optimisation accuracy would be statistically meaningless.

Algorithm 1 Two-phase Benders with partial decomposition and in-out stabilisation

Build master (10) with $\theta_s \geq \underline{\theta}_s$, and the Jensen block(s) of the scenario group means
Phase 1: relax integrality; $\bar{\mathbf{y}} \leftarrow$ first master solution
repeat
  $(\hat{\mathbf{y}}, \hat{\mathbf{w}}) \leftarrow$ solve LP master; LB $\leftarrow$ master objective
  $(\tilde{\mathbf{y}}, \tilde{\mathbf{w}}) \leftarrow \alpha(\hat{\mathbf{y}}, \hat{\mathbf{w}}) + (1-\alpha)(\bar{\mathbf{y}}, \bar{\mathbf{w}})$
  solve all $Q_s(\tilde{\mathbf{y}}, \tilde{\mathbf{w}})$ in parallel; add the cuts; $\mathrm{UB}_{\mathrm{LP}} \leftarrow \min(\mathrm{UB}_{\mathrm{LP}}$, value at $\tilde{\mathbf{y}})$; $(\bar{\mathbf{y}}, \bar{\mathbf{w}}) \leftarrow (\tilde{\mathbf{y}}, \tilde{\mathbf{w}})$
until $(\mathrm{UB}_{\mathrm{LP}} - \mathrm{LB})/\mathrm{UB}_{\mathrm{LP}} \leq \epsilon_{\mathrm{LP}}$
Phase 2: deactivate the Jensen rows, restore integrality, $\epsilon_0 \leftarrow 2 \cdot 10^{-3}$
repeat
  $(\hat{\mathbf{y}}, \hat{\mathbf{w}}) \leftarrow$ solve master MIP to gap ϵ_k (MIP start: incumbent); LB $\leftarrow \max(\mathrm{LB}$, dual bound)
  solve all $Q_s(\hat{\mathbf{y}}, \hat{\mathbf{w}})$ in parallel; update incumbent and UB; add violated cuts
  if no cut was violated, tighten ϵ_k
until $(\mathrm{UB} - \mathrm{LB})/\mathrm{UB} \leq \epsilon$

Table 4: Extensive form (HiGHS MIP, 1800 s limit) vs. Benders (Algorithm 1) with 1 and 24 subproblem workers. Speed-up: extensive form vs. Benders with 24 workers.

	Extensive form (HiGHS MIP)			Benders				Speed-up
N	columns	time (s)	gap (%)	1 worker (s)	24 workers (s)	gap (%)	iter.	
5	12,805	3	0.008	14.9	15.0	0.050	45	0.2×
10	25,480	6	0.016	4.0	9.7	0.041	33	0.6×
25	63,505	23	0.015	7.0	4.0	0.048	20	5.7×
50	126,880	48	0.042	24.2	16.6	0.028	18	2.9×
100	253,630	174	0.037	43.2	14.6	0.039	17	12×

5.5 Benchmarks for the planning paradigm

Besides the stochastic program (SP), two deterministic planning paradigms are evaluated.

- **Deterministic (median).** Plan for a single point forecast, the median, using the same model with one scenario. This is the mean-value problem whose out-of-sample cost defines the value of the stochastic solution (VSS).
- **Deterministic (P90).** Plan for the 90th percentile of every series. This is a common safety-buffer heuristic. It ignores that series do not peak together.

6 Computational study

All timings are wall-clock on a 28-core workstation, with no other experiment running, to a relative optimality gap of $5 \cdot 10^{-4}$. Scenarios are drawn from the true demand distribution of a regular day unless stated otherwise.

Extensive form vs. decomposition. Table 4 compares the deterministic equivalent solved directly by HiGHS with the decomposition. HiGHS is a strong MIP solver and handles the extensive form well for few scenarios: at $N = 5$ it is five times faster than Benders. Its solve time grows superlinearly with the number of scenarios, however: 47 s at $N = 50$ and 174 s at $N = 100$, with 254,000 columns. Benders' solve time, in contrast, barely moves. The decomposition wins from $N = 10$ with one worker and from $N = 25$ with 24 workers, where pool overhead is amortised. At $N = 100$ it is 12× faster with 24 workers (4× with one).

Scaling to many scenarios and parallelism. With 24 workers, Benders solves $N = 1000$ (2.5 million recourse columns) in 41 s (Figure 5a). The number of iterations is essentially independent of N (16–18). The phase-1 Jensen block gives the master a scenario-independent description of

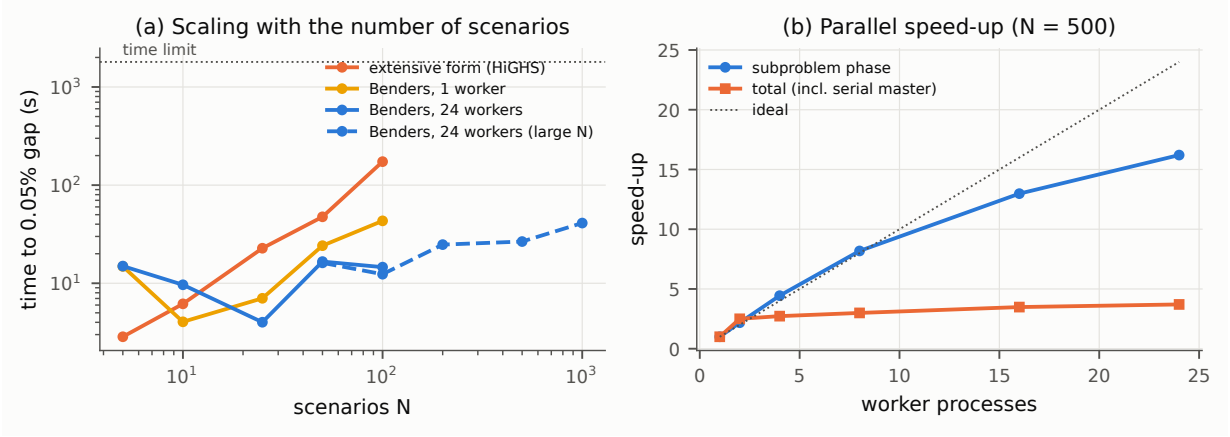


Figure 5: (a) Time to a 0.05% gap as a function of the number of scenarios. (b) Parallel speed-up for $N = 500$: the subproblem phase scales well, the total is limited by the serial master problem (Amdahl's law).

Table 5: Ablation of the Benders components ($N = 50$, 24 workers, three regular test days, 900 s limit). Each row adds one component to the row above, up to "+Jensen (phase 1)", which is the configuration used everywhere else. The last three rows are variations of it. Master: share of the time spent in the master problem. †Time limit reached on at least one day.

Variant	mean time (s)	max time (s)	iterations	LP iter.	cuts	max gap (%)	master (%)
classic†	927.0	963.7	95	0	4750	0.076	100
+LP phase	53.5	90.5	88	86	4416	0.045	97
+in-out	38.2	47.6	52	49	2616	0.037	97
+Jensen (phase 1)	9.9	11.9	18	15	882	0.042	96
+Jensen (both phases)	10.7	22.9	18	15	882	0.039	96
+Jensen x4 (phase 1)	7.6	12.3	13	10	633	0.042	96
single-cut (all)†	905.3	908.6	346	150	346	0.113	99

the capacity structure, and multi-cut lets every scenario contribute its own cut in each iteration. Parallel subproblems scale well: the subproblem phase at $N = 500$ is 16× faster with 24 workers than with one (Figure 5b). The total speed-up, however, is only 3.7×, because the master MIP is solved serially and accounts for over 90% of the time with many workers. The natural next step is therefore not more workers but a single-tree *branch-and-Benders-cut*, in which cuts are added as lazy constraints inside one branch-and-bound search instead of re-solving the master MIP in every iteration.

Ablation. Table 5 and Figure 6 isolate the contribution of each component.

- **Classic multi-cut Benders** reaches the 0.05% target on only one of the three days within 900 s. In its first ten iterations the cuts carry almost no information about the capacity structure and the lower bound stays flat (Figure 6b).
- **The LP phase** is the largest single gain: it generates most cuts from cheap master LPs, and the time falls to 54 s on average.
- **In-out stabilisation** reduces the number of iterations by 40% and the mean time to 38 s. It is not uniformly better: on one of the three days it is slower than the LP phase alone (48 s vs. 33 s).
- **The Jensen block** gives the master a valid, scenario-independent approximation of the recourse from the first iteration. The first lower bound is within 1.2% of the optimum, the iterations drop to 18 and the time to 10 s. In total the method is at least 90× faster than classic Benders; "at least" because classic Benders hit the time limit on two days.
- **Variations.** Keeping the Jensen block in the integer phase does not help on average and is

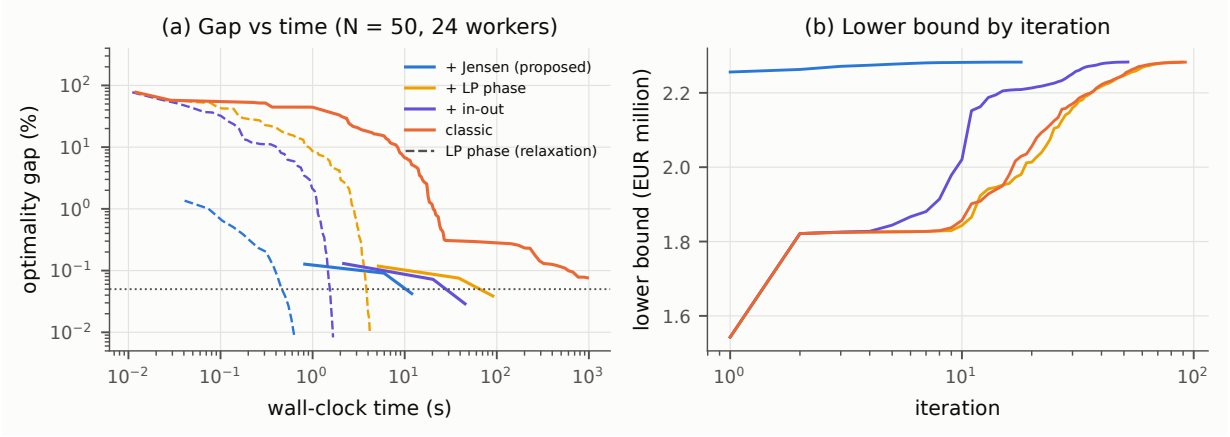


Figure 6: Convergence of the ablation variants on 5 March 2026. (a) Optimality gap against time. Dashed: gap of the LP relaxation during the LP phase; solid: true gap of the integer problem (best integer solution vs. best bound). Dotted line: the 0.05% target. (b) Lower bound by iteration; the Jensen block makes the first bound almost tight.

slower on one day (23 s): it adds rows to every master MIP. Four Jensen blocks instead of one save another 25% (7.6 s). We keep the simpler single block.

- **Single-cut** Benders (one aggregated cut per iteration) does not reach the target in 900 s even with all other components: aggregating 50 scenarios into one cut throws away most of the information returned by the subproblems.

In every variant the master accounts for 96–100% of the time. The subproblems, warm-started and solved in parallel, are not the bottleneck; this again points to branch-and-Benders-cut as the next step.

Larger networks. All results above are for the Iberian network (130 integer variables). On a synthetic network of the same size (10 FCs, 12 SCs, 65 zones, $N = 50$), Benders behaves as on the real one: 26 s and 22 iterations. On a synthetic network twice as large (20 FCs, 25 SCs, 150 zones, 520 integer variables), it did not reach the 0.05% target within an hour, and we stopped the experiment. Since the subproblems are not the bottleneck, this is the serial master MIP again: scaling to national networks with hundreds of sites needs branch-and-Benders-cut, or a master that is itself decomposed by region. We list this as the main computational limitation (Section 11).

7 Planning results

Figure 7 shows a typical day-ahead plan. Most trucks run on short, next-day-feasible lanes out of the large Madrid, Barcelona and Seville sites; local lanes, from an FC to the sort centre of its own city, are hidden under the markers. The service map reveals the structural weakness of the network. Galicia and especially northern Portugal can be reached overnight only from the small Asturias site, so a large part of their demand is delivered on day two. We come back to this in Section 10.

7.1 Statistical quality of the SAA solutions

How many scenarios are enough? For a regular day (5 March 2026) and for Black Friday 2025 we solve $M = 10$ independent SAA replications for each $N \in \{10, 25, 50, 100, 200, 400\}$, with scenarios from the true distribution. Every resulting plan is evaluated on the same 5000 fresh scenarios. Following Mak et al. [21]:

- the mean of the M SAA optimal values is a statistical *lower* bound on the true optimum (in expectation), with a t-based confidence interval;

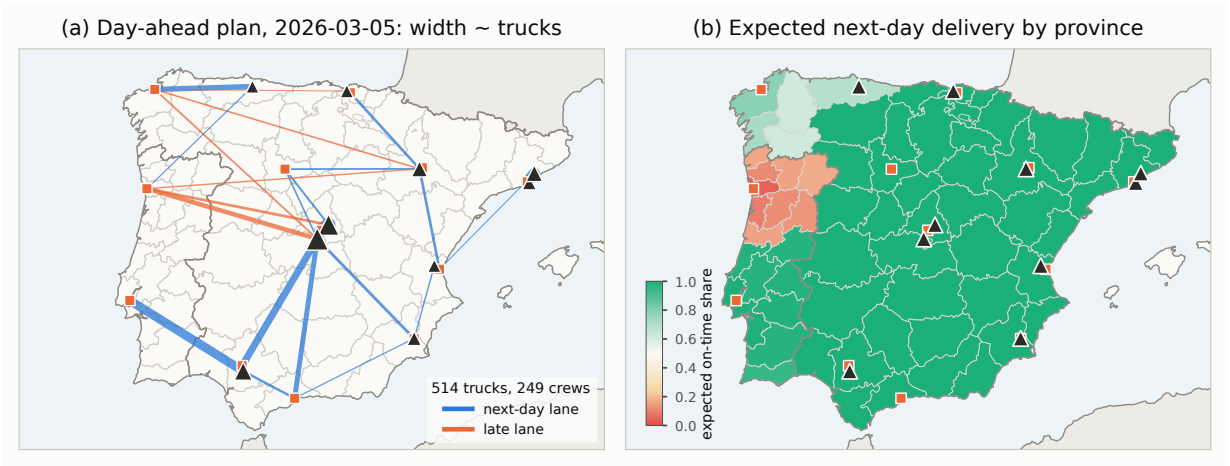


Figure 7: (a) Day-ahead SP plan for 5 March 2026 with LightGBM+copula scenarios: line width is proportional to the number of scheduled trucks per lane (blue: next-day lane; orange: late lane); FC marker size is proportional to the scheduled crews. (b) Expected share of next-day deliveries per province under this plan, over 300 true demand scenarios.

Table 6: SAA bounds ($M = 10$ replications per N, out-of-sample evaluation on 5000 true scenarios). Gap CI: 95% upper confidence bound on the optimality gap of the best plan, $(\text{UB} - \text{LB} + \text{hw})/\text{UB}$. Time: mean Benders solve time (24 workers).

Day	N	Lower bound (95% CI)	Best UB	Mean UB	Gap (%)	Gap CI (%)	Time (s)
2025-11-28	10	4,609,718 ± 61,195	4,609,066	4,611,047	-0.014	1.314	0.5
	25	4,608,203 ± 28,740	4,609,334	4,609,642	0.025	0.648	0.4
	50	4,609,204 ± 15,447	4,608,596	4,609,259	-0.013	0.322	0.7
	100	4,608,588 ± 16,167	4,607,959	4,609,119	-0.014	0.337	1.2
	200	4,611,299 ± 15,916	4,608,849	4,609,101	-0.053	0.292	1.9
	400	4,604,010 ± 9,490	4,608,717	4,608,990	0.102	0.308	3.9
2026-03-05	10	2,288,673 ± 27,928	2,290,941	2,292,870	0.099	1.318	11.8
	25	2,288,738 ± 13,196	2,290,634	2,291,043	0.083	0.659	7.0
	50	2,289,886 ± 6,882	2,290,485	2,290,697	0.026	0.327	14.9
	100	2,289,350 ± 7,560	2,290,383	2,290,584	0.045	0.375	17.5
	200	2,291,186 ± 7,021	2,290,276	2,290,469	-0.040	0.267	30.2
	400	2,288,141 ± 4,345	2,290,262	2,290,520	0.093	0.282	30.4

- the best out-of-sample value is an *upper* bound.
  Table 6 and Figure 8 show that the planning problem is statistically well behaved.
- The true expected cost of the SAA plans stabilises quickly. Its spread across replications falls from €2239 per day at $N = 10$ to about €140 at $N \geq 50$, and the plans at $N = 100$ and $N = 400$ differ by only €64 per day.
- The estimated optimality gap is statistically indistinguishable from zero for all $N \geq 25$. Its 95% upper confidence bound falls to 0.3–0.4% at $N \geq 50$, a level limited by the variance of the lower-bound estimator, not by the quality of the plans.

We therefore use $N = 100$ throughout. Note also the optimisation tolerance of $5 \cdot 10^{-4}$ (Section 5.4): it is of the same order as these statistical errors, so solving more tightly would be pointless.

Value of the stochastic solution and of perfect information. Table 7 decomposes the value of information.

- **VSS.** The stochastic plan saves €3.2k per day on the regular day and €7.1k on Black Friday over the plan for mean demand. This is small (0.14% and 0.15%) but highly significant.

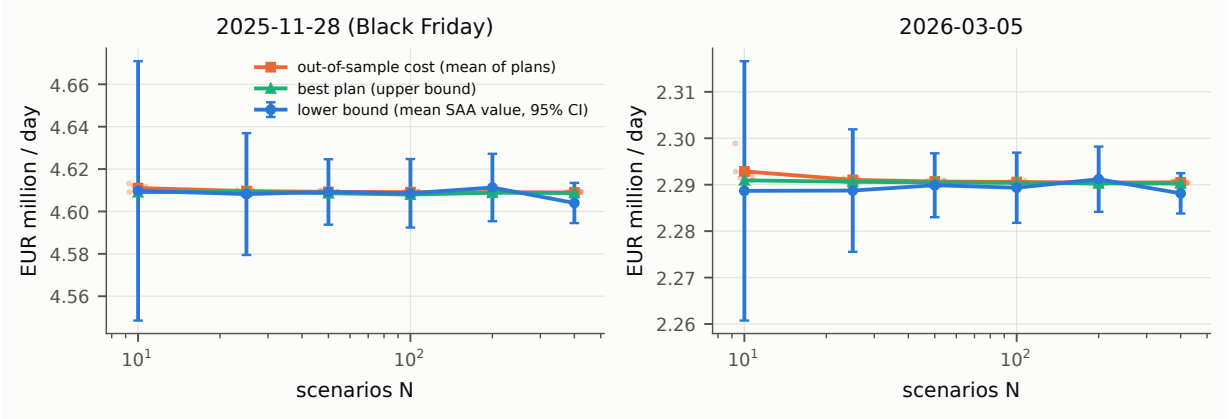


Figure 8: SAA convergence. The plans' true expected cost (orange) stabilises from $N \approx 50$; the lower-bound estimator (blue, 95% CI) tightens around it as N grows.

Table 7: Value of the stochastic solution (VSS = EEV − RP, paired over 5000 scenarios, ± 95% CI) and expected value of perfect information (EVPI = RP − WS over 100 scenarios). RP: stochastic plan; EEV: plan for the mean demand; WS: wait-and-see.

Day	RP €/day	EEV €/day	VSS €/day	WS €/day	EVPI €/day	Unmet % RP / EEV	Trucks RP / EEV	Crews RP / EEV
2026-03-05	2,290,262	2,293,459	3,196 ± 159	2,273,811	22,698 ± 2,602	0.12 / 0.03	519 / 530	252 / 257
2025-11-28	4,607,959	4,615,030	7,071 ± 230	4,607,093	18,130 ± 1,590	0.79 / 0.77	954 / 952	425 / 425

- **EVPI.** Perfect information would be worth €18–€23k per day, six to seven times more. Better *information*, i.e. forecasts, matters more than better *modelling of uncertainty*, consistent with Section 7.3.
- **Direction of the hedge.** The SP plan buys *fewer* trucks (519 vs. 530) and crews (252 vs. 257) than the mean-value plan. This is the newsvendor logic. Scheduled capacity costs c and its same-day substitutes cost $1.6c$ (overtime) and $1.8c$ (spot), so the critical ratio $(1.8c - c)/(1.8c) \approx 0.44$ is below one half. It is optimal to plan *below* the mean and buy flexibility on the day. A planner who reasons only in terms of point forecasts would build the opposite safety margin. On Black Friday the labour limits bind (425 of 426 crews), so the hedge acts on trucks only.

7.2 When does stochastic planning pay?

The modest VSS of the base case is a property of the cost structure, not of the method. Same-day substitutes for scheduled capacity are only 60–80% more expensive, and overtime can add 30%. We therefore vary the *flexibility of recourse* on four regular days spread over the year:

- the spot premium, from 1.4× to 3.5×;
- the overtime cap, 30% or 10% of planned labour;
- the penalty for an unshipped unit, from €9 to €50.

For each setting we compute the VSS and the reduction of the 95% conditional value-at-risk (CVaR_{95}, the mean of the worst 5% of days) on the true distribution. In a second experiment (10 regular days) we repeat the forecast-based comparison of Section 7.3 under base and inflexible recourse.

Table 8 and Figure 9 give a clear answer.

1. **The value of stochastic planning grows with the cost of same-day flexibility.** The VSS triples, from €2.6k to €7.7k per day, between the base case and the inflexible case (spot 3.5×, overtime cap 10%). The protection of the bad tail grows much faster: the SP plan lowers the CVaR_{95} of the daily cost by €47k per day in the inflexible case, against €2k in the base case. When recourse is expensive, the stochastic plan is mainly an insurance against bad days.

Table 8: Value of the stochastic solution as a function of recourse flexibility (means over four regular test days; SP and mean-value plans evaluated on 1000 true scenarios). ΔCVaR_{95}: reduction of the expected cost of the worst 5% of days achieved by the SP plan.

Spot premium	Overtime cap (%)	Unmet penalty	VSS €/day	95% ±	VSS %	Trucks RP / EEV	Crews RP / EEV	Unmet (%) RP / EEV	ΔCVaR_{95} €/day
1.4	10	15	4,382	199	0.17	553 / 576	284 / 286	0.09 / 0.09	19,216
1.4	30	15	3,221	253	0.13	552 / 576	282 / 286	0.06 / 0.03	-3,564
1.8	10	15	3,654	203	0.15	569 / 578	284 / 286	0.14 / 0.12	19,666
1.8	30	9	3,256	169	0.13	535 / 550	272 / 278	1.81 / 1.56	1,998
1.8	30	15	2,552	164	0.10	572 / 578	282 / 286	0.10 / 0.05	1,835
1.8	30	25	2,155	163	0.08	572 / 578	282 / 286	0.00 / 0.00	2,900
1.8	30	50	2,183	209	0.08	571 / 578	282 / 286	0.00 / 0.00	5,257
2.5	10	15	4,665	319	0.18	583 / 580	284 / 286	0.18 / 0.18	33,853
2.5	30	15	3,435	161	0.14	582 / 580	282 / 286	0.15 / 0.10	10,690
3.5	10	15	7,662	459	0.30	592 / 580	283 / 286	0.25 / 0.27	46,556
3.5	30	15	5,688	262	0.23	592 / 580	282 / 286	0.20 / 0.19	25,113

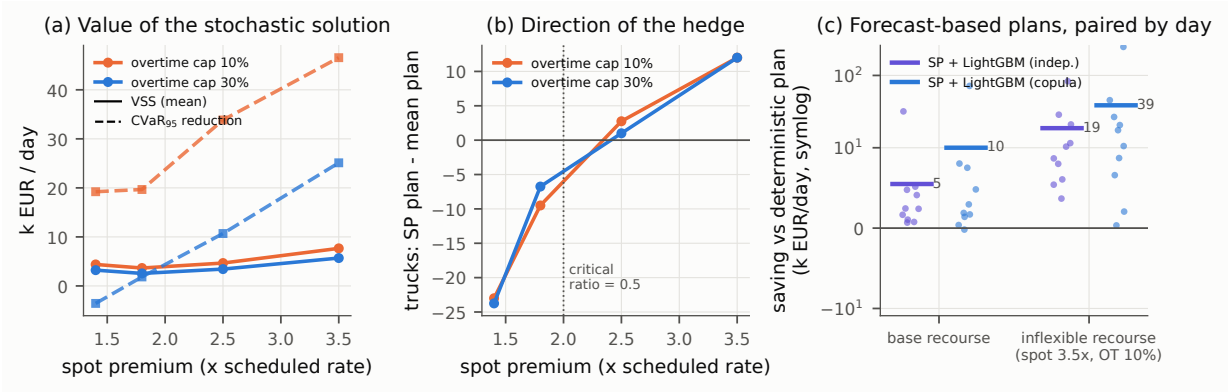


Figure 9: (a) VSS (solid) and CVaR_{95} reduction (dashed) against the spot premium, for two overtime caps. (b) Difference between the trucks planned by the SP and by the mean-value plan: the hedge changes sign where the newsvendor critical ratio crosses one half. (c) Daily saving of the forecast-based SP plans over the deterministic median plan, paired by day (bars: means).

2. **The direction of the hedge follows newsvendor theory** (panel b). If c is the per-unit price of scheduled capacity and γc that of its same-day substitute, under-provisioning costs $(\gamma - 1)c$ per unit and over-provisioning c. The critical ratio $(\gamma - 1)/\gamma$ is therefore below one half when $\gamma < 2$. In that regime the SP buys *fewer* trucks than the mean-value plan (up to 24 fewer at $\gamma = 1.4$); above it, it buys *more* (12 more at $\gamma = 3.5$). A planner who always adds a safety margin is right only in the second regime.
3. **Joint calibration pays when recourse is expensive** (panel c). With real LightGBM forecasts, the SP plans beat the deterministic median plan by €5k (independent scenarios) and €10k per day (copula) under base recourse. Under inflexible recourse they do so by €19k and €39k per day, on all 10 days. The copula's advantage over independent scenarios grows from €5k to €20k per day (better on 8 of 10 days). This resolves the null result of Section 7.3. Dependence modelling matters exactly when the network cannot cheaply absorb a correlated, network-wide surprise.
4. **The unshipped-unit penalty** has little effect on the VSS once it exceeds the cost of overtime plus spot (€15–€50). At €9 (Section 3) leaving 1.6–1.8% of demand unshipped becomes the cheapest recourse, which we consider an unrealistic operating regime for a next-day network.

Table 9: Planning regret of forecast/paradigm combinations, relative to the SP with the true demand distribution (expected cost over 2000 true scenarios per day). Standard errors are across days. Mean expected daily cost of the reference: €2.84 million.

Plan	Regular days ($n = 24$)			Events ($n = 7$)	All days (means)				
	Regret €/day	s.e.	%	Regret €/day	On-time %	Unmet %	Trucks	Crews	Time s
Deterministic (P90)	121,669	11,256	4.91	81,190	91.47	0.37	702	339	2.7
Deterministic (median)	19,277	5,776	0.76	23,100	91.40	0.51	598	290	8.3
SP + seasonal naive	43,799	6,529	1.77	1,184,378	88.58	3.95	527	278	9.9
SP + LightGBM (indep.)	16,228	4,829	0.64	20,586	91.51	0.52	597	290	21.9
SP + LightGBM (copula)	14,019	3,505	0.57	24,138	91.55	0.56	590	286	8.6
SP + true distribution	0	0	0.00	0	91.61	0.46	604	293	11.8

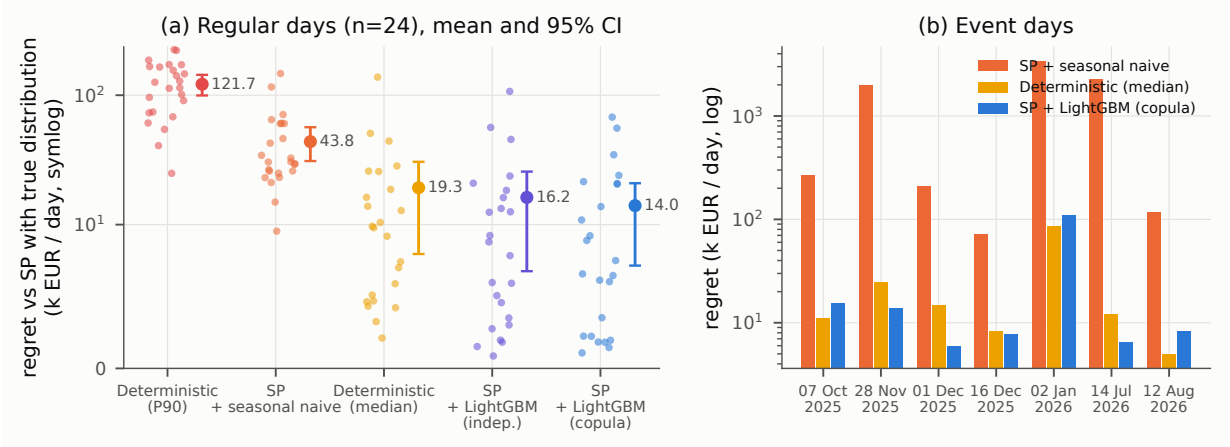


Figure 10: (a) Daily regret on regular days (dots) with mean and 95% confidence interval (symlog scale). (b) Event days: a forecast that does not know the retail calendar (seasonal naive) leaves millions of euros on the table on Black Friday, Prime Day and the Reyes peak.

7.3 What is a better forecast worth? Decision-focused evaluation

We now close the loop between Section 4 and the planner. On each of the 31 planning test days (24 regular days and 7 event days spread over the test year) we build first-stage plans from different forecasts and planning paradigms. Every plan is then scored on the *same* 2000 draws from the true conditional demand distribution (common random numbers), by solving the recourse LP for each draw. The reference is the stochastic program solved with scenarios from the true distribution ("SP + true distribution"); the regret of any other plan is the extra expected daily cost it causes. Unlike a forecast metric, this measures what operations actually pay.

Table 9 and Figure 10 support four conclusions.

1. **Forecast quality dominates.** Planning with LightGBM instead of the true distribution costs €16.3k per day on average (paired standard error €4.2k; worse on all 31 days). This is about 0.6% of the daily cost, or roughly €6M per year at this volume, and it is an upper bound on what any further forecasting improvement could deliver. The seasonal-naive forecast has a regret of €44k per day on regular days (€30k more than LightGBM), and on event days it is catastrophic (€1.2M per day on average): it schedules the capacity of an ordinary Friday for Black Friday.
2. **Safety buffers are expensive.** Planning every series at its 90th percentile, a common heuristic, ignores that the 260 series do not peak together. It over-buys about 110 trucks and 50 crews per day and costs €96k per day more than the stochastic program (paired $p < 10^{-8}$), i.e. about €35M per year.
3. **The stochastic program beats the deterministic plan, modestly.** With the same

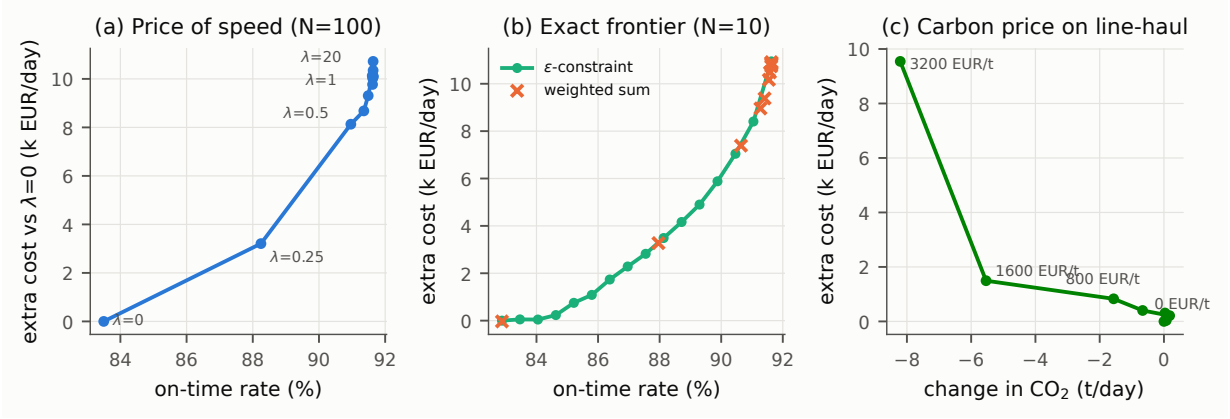


Figure 11: (a) Out-of-sample cost–speed frontier from the weighted-sum sweep. (b) Exact in-sample frontier from the ε-constraint method (16 points) and the weighted-sum solutions on the same 10-scenario SAA. The weighted sum clusters at the ends; the ε-constraint covers the frontier evenly. (c) Cost–CO_2 trade-off under a carbon price on line-haul trucks.

LightGBM forecast, the SP with independent scenarios saves €2.9k per day over the median plan (paired s.e. €1.1k, $p = 0.013$, better on 74% of days). The SP with copula scenarios saves €3.8k per day (s.e. €2.5k, $p = 0.14$). The gain is small because recourse is cheap in the base case; Section 7.2 shows when it becomes large.

4. **Correct joint calibration does not automatically pay.** The copula repairs the calibration of the national total (Table 3), but the paired cost difference to independent scenarios (€0.9k per day) is not significant. The lesson is general: when recourse prices are nearly linear around the optimum, the plan is driven by the *mean and the marginal quantiles at the critical ratio*, not by the shape of the joint distribution. Better forecasts of the level beat better forecasts of the dependence.

8 Multi-objective analysis: cost, speed and CO_2

The late penalty λ is not a cost that appears on an invoice. It is a *policy choice* about how much the business is willing to pay for the next-day promise. Rather than fixing it, we trace the trade-off for a representative regular day (5 March 2026). All plans are evaluated out of sample on 2000 true scenarios. Each plan is operated with the price of speed it was planned with, so the recourse assignment is consistent with the plan.

Methods.

1. **Weighted sum**: solve the SAA ($N = 100$, Benders) for 13 values of $\lambda \in [0, 20]$ €/late unit.
2. **Augmented ε-constraint**, on a 10-scenario SAA solved exactly as an extensive form. Minimise cost subject to $\sum_s p_s(\text{late}_s + \text{unshipped}_s) \le \varepsilon \sum_s p_s D_s$, for 16 equally spaced values of ε between the two extremes of the weighted sum. A tiny tie-breaking penalty (10^{-3} €/unit) on lateness removes weakly efficient solutions, which is what AUGMECON's slack term achieves [23, 24]. Because the ε-constraint couples all scenarios, a plan obtained this way does not by itself say how to operate on a given day. We therefore re-solve the LP with the integer first stage fixed and read the Lagrange multiplier $\lambda^\star_\varepsilon$ of the ε-row. By LP duality, a recourse that prices lateness at $\lambda^\star_\varepsilon$ reproduces the planned service level; we verified this numerically to five digits. $\lambda^\star_\varepsilon$ is also the *marginal price of the last point of speed* at that point of the frontier.
3. **Carbon price**: add κ €/t CO_2 to the line-haul cost per truck-km, for κ up to €3200/t.

Findings.

1. **Speed is cheap, until it is impossible.** Moving from the cheapest plan ($\lambda = 0$, 83.5% on time) to $\lambda = 0.5$ (91.0%) costs only €8.1k per day (0.36%). The frontier then turns vertical at 91.6%. The last 8.4% of units cannot be delivered next day at any price on this day: their zones are reachable on time only from FCs that lack the inventory or capacity (Figure 7). This is a *network design* limit, not an operational one; Section 10 prices the investment that moves it.
2. **Speed is bought in the assignment layer.** Across the whole frontier the plan keeps 516–523 trucks and 251–253 crews. Faster plans route differently through the same capacity rather than buying more. This is useful for leadership communication: the promise level is mostly an *operational* decision that can be changed daily, without re-sizing the network.
3. **Marginal prices along the frontier.** The multipliers $\lambda^\star_\varepsilon$ grow from €0.003 to €0.42 per late unit up to 90% on time, and reach €1.98 for the last attainable point. The default $\lambda =$ €1 therefore sits at the knee of the frontier.
4. **Weighted sum vs. ε-constraint.** With integer trucks the frontier could in principle be non-convex, with unsupported points that no weighted sum can reach. We tested this against the lower convex hull of all 29 exact solutions. Eight ε-points lie above the hull, but by at most €125 on €2.27M, which is below the MIP tolerance. The frontier is therefore convex for practical purposes. The value of the ε-method here is *coverage*: seven of the 13 weighted-sum solutions collapse onto practically the same point at $\lambda \geq 2$, whereas the ε-constraint delivers evenly spaced options, each with its marginal price.
5. **Line-haul carbon is expensive to abate.** A carbon price below €400/t changes nothing: trucks are already well consolidated, and the lanes are fixed by geography. At €1600–€3200/t the plan sheds 10–23 trucks and cuts 5.5–8.2 t of CO_2 per day (6–9% of line-haul emissions of about 92 t; last mile adds 77 t that the plan cannot influence). It does so by consolidating volume onto fewer, fuller trucks and shifting the overflow to spot capacity. This costs €1.5–€9.5k per day, an implied abatement cost of €270–€1160/t before counting the 1.3–2.2 points of on-time delivery that are lost. Speed itself has a carbon cost: the next-day plan ($\lambda = 1$) emits 7.3 t (4.5% of total, 8% of line-haul) more CO_2 than the cheapest plan ($\lambda = 0$), because on-time lanes are often longer ones.

9 Online order assignment under the day-ahead plan

The day-ahead plan fixes the capacities: Qy_{ih} package-equivalents of scheduled lane capacity, πw_i regular labour units, and an overtime allowance of $\rho\pi w_i$. During the day, units arrive and must be assigned *irrevocably*, without knowing the rest of the day. This is the decision Amazon-style controllers take "millions of times per hour". The recourse LP of Section 5 evaluated on the realised demand is the *hindsight* benchmark. It is a lower bound on the cost of any non-anticipative policy that uses the same plan, because every online trajectory is a feasible solution of that LP.

Simulation model. The realised daily demand $\mathbf{d}$ (a draw from the true distribution) is split into 96 quarter-hour slots by a Dirichlet-perturbed intraday profile, with a night trough and lunch and evening peaks. Within a slot, the batches (j,k) arrive in random order. Assigning one unit of (j,k) to FC i consumes one unit of Inv_{ik}, l_k labour units and v_k package-equivalents of lane $(i,h(j))$. Regular labour is consumed first, then overtime (at c^{ot}, up to the cap); scheduled truck space is consumed first, then spot space (at s_{ih}). The *immediate* cost of a unit is therefore

$$\gamma_{ijk} = c^{\mathrm{fc}}_i + c^{\mathrm{lm}}_j + \lambda\ell_{ij} + c^{\mathrm{ot}} l_k \mathbb{1}[\text{regular labour at } i \text{ exhausted}] + s_{i,h(j)} v_k \mathbb{1}[\text{lane } (i,h(j)) \text{ full}].$$

A unit with no feasible FC (no inventory or no labour left anywhere it is stocked) is left unshipped at cost $p + \lambda$. Batches are split exactly at the points where a resource changes cost regime, so the simulation is equivalent to unit-by-unit processing within a batch.

Table 10: Online assignment over 124 simulated days (31 test days × 4 replications) under the day-ahead SP plan. Regret is the recourse cost above the clairvoyant hindsight LP; "gap closed" is the share of greedy's regret removed. Spot and overtime are daily means.

Policy	Recourse cost €/day	Regret %	s.e.	Gap closed %	On-time %	Unmet %	Spot pkg-eq	Overtime labour u.	LPs
Greedy	2,300,479	7.10	0.41	0	87.52	1.08	146,666	54,897	0
Bid price (static)	2,273,493	5.56	0.46	22	88.42	1.07	123,145	58,348	1
Bid price (re-solve)	2,205,156	2.35	0.45	67	90.37	0.89	77,349	48,941	24
Fluid LP tracking	2,190,863	1.53	0.48	79	90.94	0.91	64,586	48,982	24
Hindsight LP	2,116,083	0.00	0.00	100	91.54	0.61	47,906	46,302	1

Policies.

- **Greedy**: $i^\star = \arg\min_i \gamma_{ijk}$. This is myopic: it depletes the cheap, on-time capacity near the early customers and leaves nothing for later ones.
- **Static bid prices** [1, 32]: at the start of the day, solve the fluid LP, i.e. the recourse LP with the full capacities and the forecast demand $\hat{\mathbf{d}}$. Take the shadow prices $\mu \geq 0$ of inventory, regular labour, overtime and lane capacity, and assign to

$$i^\star = \arg\min_i \ \gamma_{ijk} + \mu^{\text{inv}}_{ik} + l_k\, \mu^{\text{lab/ot}}_i + v_k\, \mu^{\text{lane}}_{i,h(j)},$$

  i.e. immediate cost plus the opportunity cost of the resources consumed. The labour price is the regular or the overtime one, whichever is currently being consumed; lane prices apply while scheduled space remains.
- **Re-solved bid prices**: the fluid LP is re-solved every hour with the *remaining* capacities and the *remaining* expected demand. The latter is the forecast times the remaining share of the intra-day profile, scaled by an online learning factor per category, (observed $+a$)/(expected so far $+a$) with $a = 2000$ units of shrinkage.
- **Fluid LP tracking**: with the same hourly re-solves, the units of a batch are split across FCs in proportion to the LP flows $x^\star_{ijk}$, with a greedy fallback when a target FC has run out. This is the fluid-tracking idea behind probabilistic and correlated-rounding schemes [15].
- **Hindsight**: the clairvoyant LP (lower bound).

The online experiment uses the day-ahead plans of the proposed pipeline (SP with LightGBM + copula scenarios) on all 31 test days. It runs 4 demand/arrival replications per day, i.e. 124 simulated days per policy, and the policies see the same arrivals (common random numbers).

Results. Table 10 and Figure 12 show a clear ranking.

- **Greedy assignment costs 7.1% more than hindsight**, about €184k per day. Most of the loss comes from spot capacity (three times the hindsight volume) and late deliveries. Greedy fills the cheap next-day lanes with the first customers of the day; later customers in other regions then need spot trucks or late lanes.
- **Static bid prices close 22% of this gap.** That is of the same order as the 36% reported by Acimovic and Graves [1] on industry data with a comparable dual-value heuristic. They fail when the day deviates from the forecast: on 18% of the days they are worse than greedy.
- **Re-solving hourly closes 67%**, and **fluid LP tracking closes 79%** (regret 1.5%; 0.3% on regular days). The policies are separated by €110k per day between greedy and LP tracking.
- **Robustness.** Panel (b) explains the difference between the two LP-based policies. Bid prices are *marginal* information: they are only valid locally, and degrade quickly as the state drifts from the one they were computed for (from 3.8% regret at 15-minute re-solves to 6.7% with a single solve). Tracking the LP *flows* uses the full primal plan and is almost insensitive to the re-solve frequency (3.4% to 4.0%). Multiplicative demand learning brings only small gains, because the day-ahead forecast is already conditioned on yesterday's level.

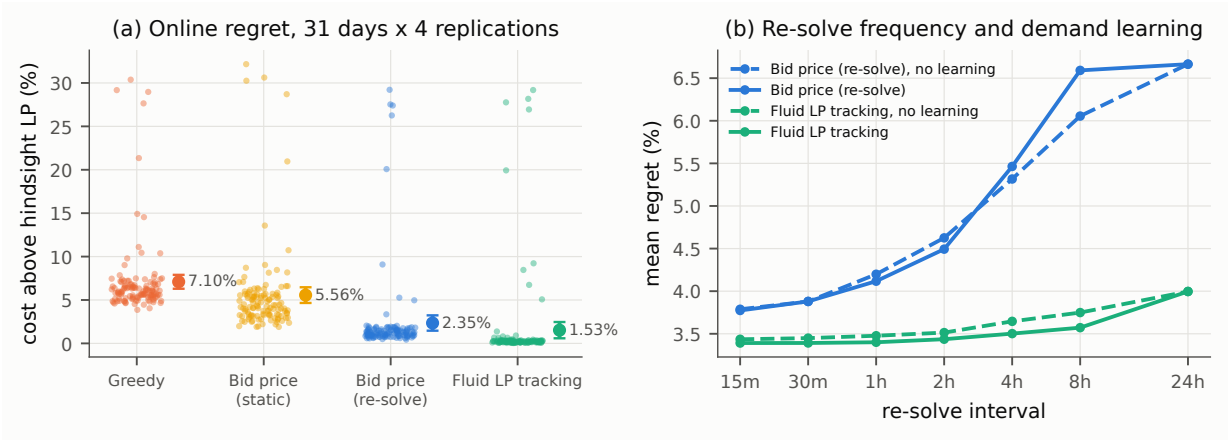


Figure 12: (a) Regret per simulated day; the outliers above 15% are event days whose demand deviates strongly from the forecast. (b) Mean regret as a function of the re-solve interval, with and without online demand learning (every third test day). LP tracking is far less sensitive to stale information than bid prices.

Table 11: Value of a fulfilment centre near Lisbon (31 test days). Savings are differences of expected daily cost (base minus extended network). "Objective" includes the price of speed λ, "cost excl. late" only out-of-pocket costs. The annualised figure is 365× the regular-day mean, i.e. conservative (events excluded). Fixed costs of the site are not included: the annualised saving is the break-even annual cost of the site.

λ	Daily saving, regular days (€)			Saving,	On-time PT	On-time all	CO_2 (t/day)	Annualised
	objective	s.e.	cost excl. late	events	(%)	(%)		(M€/yr)
1.0	50,142	1,648	25,496	141,941	53.6 → 92.6	91.61 → 96.89	202.4 → 182.9	18.30
2.5	89,843	2,830	22,717	205,465	54.4 → 96.4	91.71 → 97.47	202.9 → 185.9	32.79

- **Event days** remain the hardest, with 5.7% regret even for LP tracking: the forecast error is large and arrives early in the day. This is where intraday re-forecasting and promise-date flexibility would pay most.

Computationally, all policies run a full day of about 25,000 batch decisions in under half a second, LP solves included. The LP is the same warm-started recourse model used by the planner, re-optimised by dual simplex in milliseconds.

10 Strategic what-if: a fulfilment centre near Lisbon

The frontier of Section 8 stops at about 92% on-time delivery, and the plan map (Figure 7) shows where the missing points are. Northern Portugal (the Porto sort centre) can only be served next-day from Asturias, a small site. In practice most of that volume ships from Madrid on lanes that miss the overnight window. This is a question for the network, not for daily operations, and the same machinery can answer it.

Experiment. We add a candidate FC in Azambuja, a logistics hub about 50 km north-east of Lisbon, with 34 crews and 90 truck departures. Its lanes to both Portuguese sort centres fit the overnight window. The national inventory cover is unchanged and is re-spread over 11 FCs. On each of the 31 test days, the day-ahead SP ($N = 100$ true scenarios) is re-optimised for the base and the extended network, and both plans are evaluated on 1000 true scenarios. Using the true distribution isolates the network effect from forecast error. We report two prices of speed, because the business case depends on how much the next-day promise is worth.

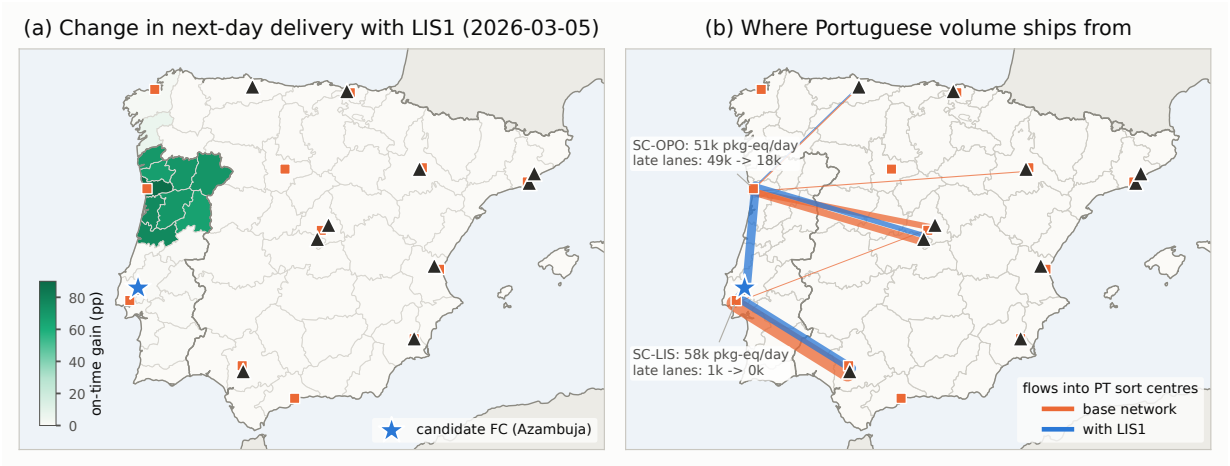


Figure 13: (a) Gain in expected next-day delivery by district when the Lisbon FC is added (5 March 2026, $\lambda = 1$): the Porto catchment gains 60–85 points; the Lisbon catchment, already served on time from Seville, is unchanged. (b) Expected package-equivalents flowing into the two Portuguese sort centres by origin FC, without (orange) and with (blue) the new site; line width is proportional to volume.

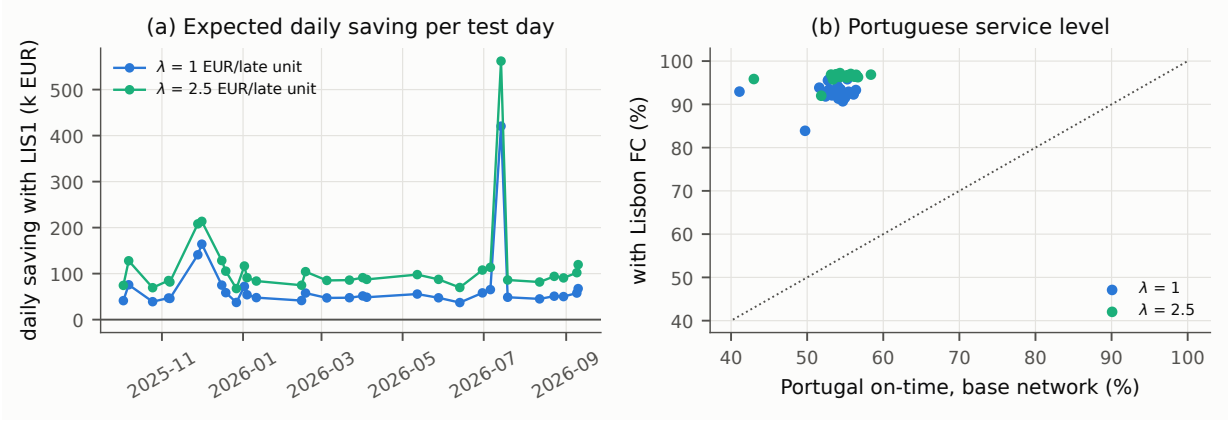


Figure 14: (a) Expected daily saving on every test day: the saving is positive on every day, and largest on peak days (Black Friday week, Prime Day), when the extra site also relieves capacity. (b) Portuguese on-time rate per test day, base vs. extended network.

Findings (Table 11, Figures 13 and 14).

- **Service.** Next-day delivery in Portugal rises from 54% to 93% ($\lambda = 1$), or to 96% ($\lambda = 2.5$). Network-wide it rises from 91.6% to 96.9%, and Spain is unaffected.
- **Cost.** On regular days, out-of-pocket operating cost falls by €25k per day, through shorter lanes, fewer spot trucks and less overtime. Including the value of speed at $\lambda = 1$, the saving is €50k per day. The saving is positive on all 31 days (minimum €37k) and peaks at €420k on Prime Day 2026.
- **Emissions.** CO_2 falls by 19.5 t per day (about 10%), because Portuguese volume no longer crosses the peninsula.
- **Break-even.** The conservative annualised saving is €9.3M per year in pure operating cost, and €18.3M per year including the value of the next-day promise at €1 per unit. A site of this size pays off if its annual fixed cost (lease, overhead, depreciation) is below these figures.

Even with the new site, part of the Porto volume still ships late from Madrid: with unchanged national cover, the inventory positioned at Azambuja is the binding constraint. That is a reminder that network design, inventory placement and capacity planning should eventually be optimised jointly (Section 11).

11 Discussion, limitations and next steps

What carries over to a production fulfilment network. The pipeline is modular in the same way production planning stacks are:

- forecasters produce quantiles;
- a scenario layer adds dependence;
- a stochastic optimiser commits capacity;
- a fast LP-dual controller assigns orders against that capacity.

Each layer can be replaced without touching the others, and each comes with its own evaluation: proper scores, joint-calibration scores, statistical optimality gaps and regret to a clairvoyant benchmark. The *decision-focused* evaluation of Section 7.3 is the piece we would most want in production. It prices forecast improvements in euros per day, which is the currency in which forecasting and planning teams should negotiate their roadmaps.

Limitations.

1. **Synthetic data.** All demand is simulated. The generative model was designed to contain the phenomena that make the real problem hard: events, holidays, promotions, weather, correlated shocks and overdispersion. It nevertheless remains a model, and real data would contain regime changes, censored demand (stock-outs) and assortment effects that it lacks. The upside of this choice is that it enables the oracle comparisons that are impossible with real data.
2. **Stylised network and costs.** Sites, capacities and costs are plausible, not real. The structural conclusions (pooling, the role of recourse flexibility, the event-day risk of naive forecasts) are robust to the exact numbers; the euro amounts are not.
3. **Single-unit shipments.** Multi-item orders introduce split-shipment costs and the correlated rounding problem of Jasin and Sinha [15], Ma [19]. The online simulator would need order-level decisions.
4. **Single-day horizon.** Unshipped units are penalised but not carried over. A rolling multi-day model, with backlog dynamics and next-day promise re-planning, would turn the two-stage model into a multi-stage one.
5. **Exogenous inventory.** Inventory placement, which ultimately determines which FCs *can* ship on time, is taken as given. Joint inventory placement and capacity planning is a natural (and harder) extension.
6. **Network size.** The decomposition is fast for the Iberian network but not yet for a network twice its size (Section 6): the serial master MIP is the bottleneck.

Next steps.

- *Forecasting:* fine-tune the foundation model with covariates; hierarchical reconciliation of the zone-level forecasts to the national level; a copula with a factor structure (national, regional and category factors) instead of a shrunk full correlation.
- *Optimisation:* branch-and-Benders-cut in a single search tree with lazy constraints; Magnanti–Wong or Papadakos Pareto-optimal cuts; a risk-averse objective (CVaR of the daily cost) through the standard Rockafellar–Uryasev reformulation, which keeps the Benders structure.
- *Learning:* a learned surrogate of the recourse function, trained on the millions of subproblem solutions the decomposition produces, to warm-start the master or screen scenarios.
- *Online control:* bid prices from a stochastic rather than a deterministic fluid LP, and multi-armed-bandit tuning of the re-solve schedule against live regret.

References

[1] J. Acimovic and S. C. Graves. Making better fulfillment decisions on the fly in an online retail environment. *Manufacturing & Service Operations Management*, 17(1):34–51, 2015.

[2] A. F. Ansari, L. Stella, C. Turkmen, X. Zhang, P. Mercado, H. Shen, O. Shchur, S. S. Rangapuram, S. Pineda Arango, S. Kapoor, J. Zschiegner, D. C. Maddix, H. Wang, M. W. Mahoney, K. Torkkola, A. G. Wilson, M. Bohlke-Schneider, and Y. Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.

[3] W. Ben-Ameur and J. Neto. Acceleration of cutting-plane and column generation algorithms: Applications to network design. *Networks*, 49(1):3–17, 2007.

[4] J. F. Benders. Partitioning procedures for solving mixed-variables programming problems. *Numerische Mathematik*, 4:238–252, 1962.

[5] J. R. Birge and F. Louveaux. *Introduction to Stochastic Programming*. Springer, 2nd edition, 2011.

[6] J. R. Birge and F. V. Louveaux. A multicut algorithm for two-stage stochastic linear programs. *European Journal of Operational Research*, 34(3):384–392, 1988.

[7] V. Chernozhukov, I. Fernández-Val, and A. Galichon. Quantile and probability curves without crossing. *Econometrica*, 78(3):1093–1125, 2010.

[8] T. G. Crainic, M. Hewitt, F. Maggioni, and W. Rei. Partial Benders decomposition: General methodology and application to stochastic network design. *Transportation Science*, 55(2): 414–435, 2021.

[9] M. Ehrgott. *Multicriteria Optimization*. Springer, 2nd edition, 2005.

[10] A. N. Elmachtoub and P. Grigas. Smart "predict, then optimize". *Management Science*, 68 (1):9–26, 2022.

[11] M. Fischetti, I. Ljubić, and M. Sinnl. Redesigning Benders decomposition for large-scale facility location. *Management Science*, 63(7):2146–2162, 2017.

[12] I. Gibbs and E. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems*, volume 34, 2021.

[13] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

[14] Q. Huangfu and J. A. J. Hall. Parallelizing the dual revised simplex method. *Mathematical Programming Computation*, 10(1):119–142, 2018.

[15] S. Jasin and A. Sinha. An LP-based correlated rounding scheme for multi-item ecommerce order fulfillment. *Operations Research*, 63(6):1336–1351, 2015.

[16] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu. LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems*, volume 30, 2017.

[17] A. J. Kleywegt, A. Shapiro, and T. Homem-de Mello. The sample average approximation method for stochastic discrete optimization. *SIAM Journal on Optimization*, 12(2):479–502, 2002.

[18] O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

[19] W. Ma. Order-optimal correlated rounding for fulfilling multi-item e-commerce orders. *Manufacturing & Service Operations Management*, 2023. arXiv:2207.04774.

[20] T. L. Magnanti and R. T. Wong. Accelerating Benders decomposition: Algorithmic enhancement and model selection criteria. *Operations Research*, 29(3):464–484, 1981.

[21] W.-K. Mak, D. P. Morton, and R. K. Wood. Monte carlo bounding techniques for determining solution quality in stochastic programs. *Operations Research Letters*, 24(1–2):47–56, 1999.

[22] S. Makridakis, E. Spiliotis, and V. Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *International Journal of Forecasting*, 38(4):1346–1364, 2022.

[23] G. Mavrotas. Effective implementation of the ε-constraint method in multi-objective mathematical programming problems. *Applied Mathematics and Computation*, 213(2):455–465, 2009.

[24] G. Mavrotas and K. Florios. An improved version of the augmented ε-constraint method (AUGMECON2) for finding the exact Pareto set in multi-objective integer programming problems. *Applied Mathematics and Computation*, 219(18):9652–9669, 2013.

[25] D. McDaniel and M. Devine. A modified Benders' partitioning algorithm for mixed integer programming. *Management Science*, 24(3):312–319, 1977.

[26] R. B. Nelsen. *An Introduction to Copulas*. Springer, 2nd edition, 2006.

[27] R. Rahmaniani, T. G. Crainic, M. Gendreau, and W. Rei. The Benders decomposition algorithm: A literature review. *European Journal of Operational Research*, 259(3):801–817, 2017.

[28] Y. Romano, E. Patterson, and E. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems*, volume 32, 2019.

[29] D. Salinas, V. Flunkert, J. Gasthaus, and T. Januschowski. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *International Journal of Forecasting*, 36(3):1181–1191, 2020.

[30] M. Scheuerer and T. M. Hamill. Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4):1321–1334, 2015.

[31] A. Shapiro, D. Dentcheva, and A. Ruszczyński. *Lectures on Stochastic Programming: Modeling and Theory*. SIAM, 2009.

[32] K. Talluri and G. van Ryzin. An analysis of bid-price controls for network revenue management. *Management Science*, 44(11):1577–1593, 1998.

[33] R. M. Van Slyke and R. Wets. L-shaped linear programs with applications to optimal control and stochastic programming. *SIAM Journal on Applied Mathematics*, 17(4):638–663, 1969.

[34] V. Vovk, A. Gammerman, and G. Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

[35] R. Wen, K. Torkkola, B. Narayanaswamy, and D. Madeka. A multi-horizon quantile recurrent forecaster, 2017. arXiv:1711.11053.

[36] P. J. Xu, R. Allgor, and S. C. Graves. Benefits of reevaluating real-time order fulfillment decisions. *Manufacturing & Service Operations Management*, 11(2):340–355, 2009.

A Notation

Symbol	Meaning
$\mathcal{I}, \mathcal{H}, \mathcal{J}, \mathcal{K}$	FCs (10), sort centres (12), demand zones (65), product categories (4)
$h(j)$	sort centre serving zone j
$\mathcal{A}$	arcs (i, j, k) with FC i stocking category k (2145)
ℓ_{ij}	1 if lane $(i, h(j))$ misses the overnight window (late delivery)
y_{ih}, w_i	first stage: scheduled trucks per lane, crews per FC
x_{ijk}, o_i, z_{ih}, u_{jk}	recourse: assignment, overtime, spot capacity, unshipped units
F_{ih}, L, Q, π	truck cost, crew cost, truck capacity, crew capacity
s_{ih}, c^{ot}, ρ	spot cost per pkg-eq., overtime cost per labour unit, overtime cap
v_k, l_k	volume and labour content of a unit of category k
p, λ	cost of an unshipped unit, cost of a late unit (price of speed)
Inv_{ik}	available-to-ship inventory
$Q(\mathbf{y}, \mathbf{w}, \mathbf{d})$	recourse value for demand $\mathbf{d}$; Q_s for scenario s
θ_s	master epigraph variable of scenario s

B Forecasting features

For origin τ, horizon $h \in \{1, \ldots, 7\}$ and series (j, k) the target is $y_{\tau+h,jk}/\bar{y}_{\tau,jk}$, where $\bar{y}_{\tau,jk}$ is the trailing 28-day mean. The features are:

- *Identifiers*: zone, category, country, h, $\log \bar{y}_{\tau,jk}$.
- *Lags* (all divided by $\bar{y}_{\tau,jk}$): $y_{\tau-l}$ for $l = 0..6$; the same weekday one and two weeks before the target; the mean of the same weekday over 8 weeks; the 7-day mean.
- *Momentum*: national 7-day/28-day ratio and last day/28-day ratio per category; regional 7/28-day ratio.
- *Event-aligned year-over-year*: last year's national and series level at the event-aligned comparable date (the same offset from Black Friday or Prime, otherwise 52 weeks earlier), relative to last year's 28-day level at the aligned origin.
- *Calendar of the target day*: weekday, month, Black Friday offset, Prime Day and Prime October indicators (and the day after), Christmas ramp and trough, New Year's Eve, Reyes ramp, *rebajas*, back-to-school, zone holiday status (holiday or day after), announced promotion lift of the category.
- *Origin context*: holiday status and event indicator of the origin day, number of event days in the scale window, promotion at the origin.
- *Weather forecast for the target day*: temperature, temperature anomaly and rain probability, with noise growing in h.

LightGBM hyper-parameters: 600 rounds, learning rate 0.05, 63 leaves, at least 400 samples per leaf, feature fraction 0.8, bagging fraction 0.7, L_2 regularisation 1. Day-of-year Fourier terms were replaced by the month after validation (Section 4).

C Reproducibility

All code is in the accompanying repository (`fulfillopt` Python package, about 2,600 lines, plus 1,700 lines of experiment scripts and unit tests). The pipeline is reproducible with `make all`. All randomness is seeded: the demand history uses seed 2026, and scenario and evaluation seeds are derived from the day index. Solvers are HiGHS 1.15.1 (via `highspy`), LightGBM 4.7 and Chronos-Bolt (small/base, zero-shot, on an RTX 5070 Ti GPU). Experiments ran on a 28-core Linux workstation with 125 GB of memory; timings in Section 6 were taken with no other experiment running.

The unit tests check:

1. that Benders cuts are valid under-estimators of the recourse function;
2. that every Benders variant reaches the extensive-form optimum within tolerance;

3. that out-of-sample evaluation reproduces the in-sample objective;
4. that quantile functions round-trip through probability integral transforms;
5. that copula sampling preserves marginals and the target correlation;
6. that every online policy conserves units, respects capacities and never beats the hindsight bound.